



sAlfer Lab

Joint lab on Safety and Security of AI

Computer Vision Technologies and Biometrics

Multimodal biometrics

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**M.Sc. Degree in Computer Engineering, CyberSecurity and
Artificial Intelligence**



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of Cagliari, Italy

Department of Electrical
and Electronic
Engineering



Outline of the talk



- **Soft biometrics**
- **Behavioural biometrics**
 - properties
 - pros and cons
 - Voice recognition
 - Gait recognition
 - Electroencephalography (EEG) recognition
- **Biometrics for people behaviour:**
 - Facial expressions
 - Gesture analysis
 - Anomaly detection in small and large crowds

Soft biometrics



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Ethnicity



Gender



Age



Eye/hair colour or hairstyle



Tattoos



- **Pros**

- Useful for improving the recognition performance
 - No personal recognition when alone!
- «Ancillary» information
- Etherogeneous insights
- «Cheap» to extract

- **Cons**

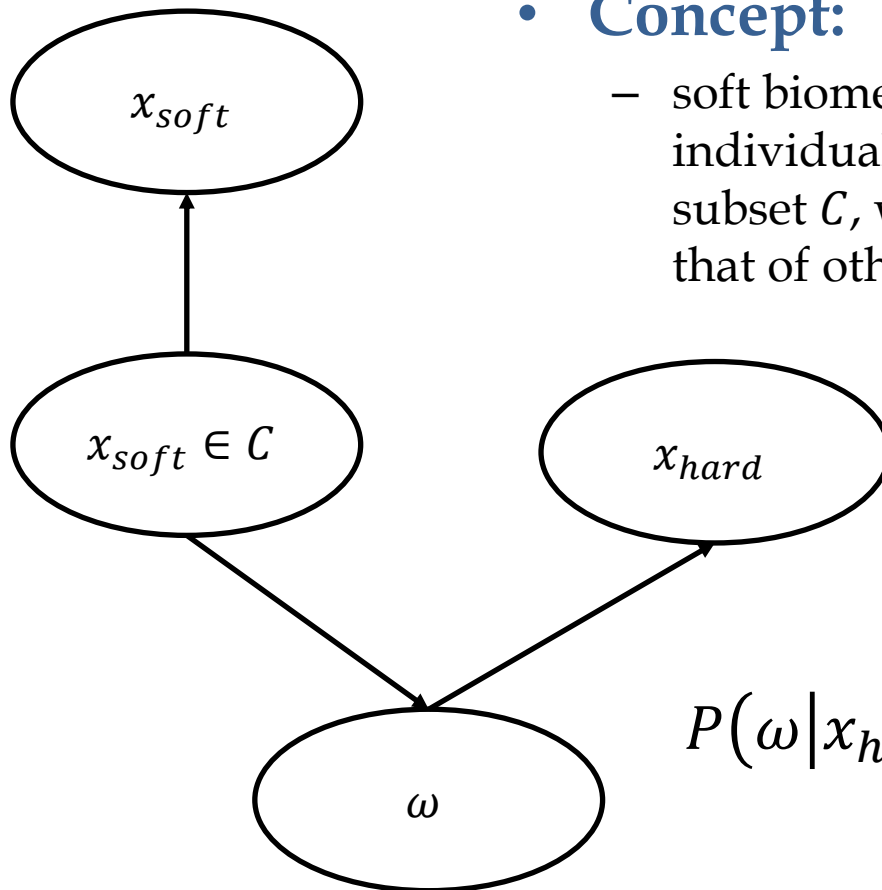
- Healthy, «behavioural» (?)
- Easy to spoof
- Difficult to classify

A Bayesian approach to integrate biometrics and soft-biometrics



- **Concept:**

- soft biometrics cannot characterize people individually, but can characterize a specific subset C , whose «size» is strongly less than that of others



Hypothesis: $P(x_{soft} \in C) = p \ll 1$

$$\exists \Omega \subset U \begin{cases} \forall \omega \in \Omega, P(x_{soft} \in C | \omega) > 0.5 \\ \forall \omega \notin \Omega, P(x_{soft} \in C | \omega) = p \end{cases}$$

$$P(\omega | x_{hard}, x_{soft} \in C) \cong a \cdot P(\omega | x_{hard})$$

$\Downarrow$

$$s = a \cdot s_{hard}; a > 1$$

How to compute a ?



- It is possible to proof that:

$$\begin{aligned} & \frac{P(\omega | x_{hard}, x_{soft} \in C)}{P(\omega | x_{hard}, x_{soft} \notin C)} = \\ &= \frac{P(x_{soft} \in C | x_{hard}, \omega)}{P(x_{soft} \notin C | x_{hard}, \omega)} \cdot \frac{P(x_{soft} \notin C | x_{hard})}{P(x_{soft} \in C | x_{hard})} = L \begin{cases} > 1, & \omega \in \Omega \\ = 1, & \omega \notin \Omega \end{cases} \end{aligned}$$

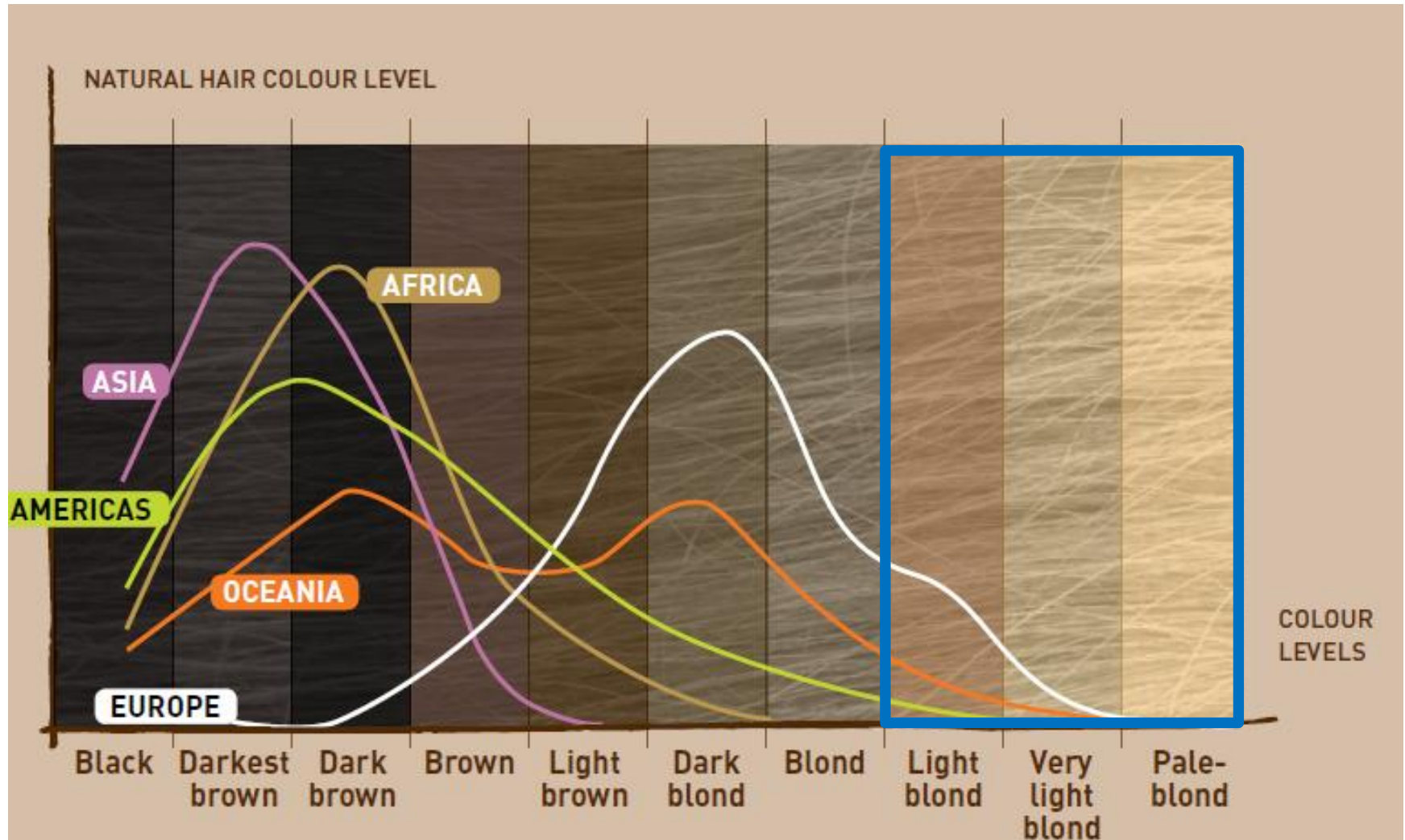
- Eventually we obtain:

$$P(\omega | x_{hard}, x_{soft} \in C) = \frac{L}{1 - p + Lp} P(\omega | x_{hard})$$

- Where:

$$a = \frac{L}{1 - p + Lp}$$

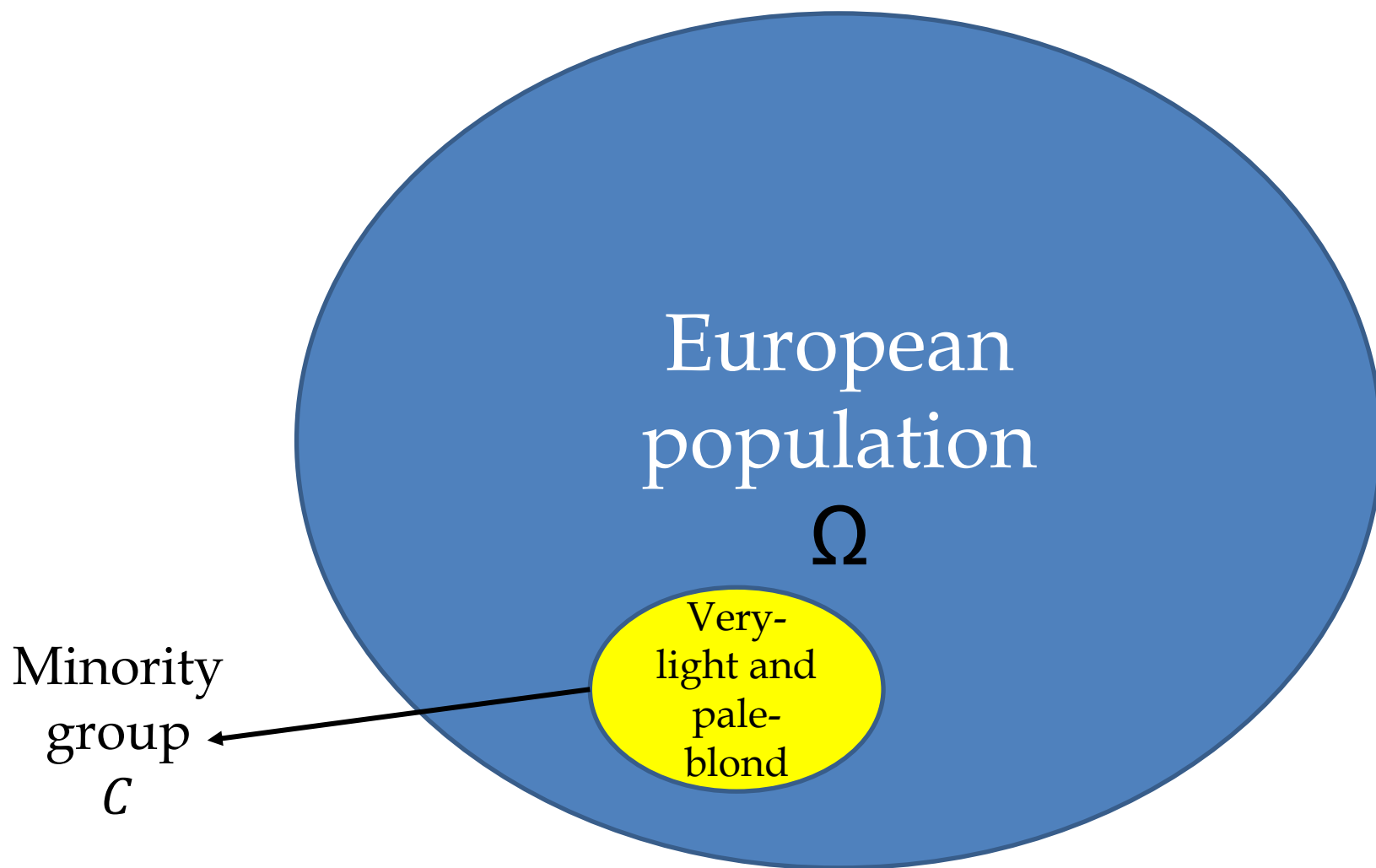
Case-study: hair colour as a soft biometric



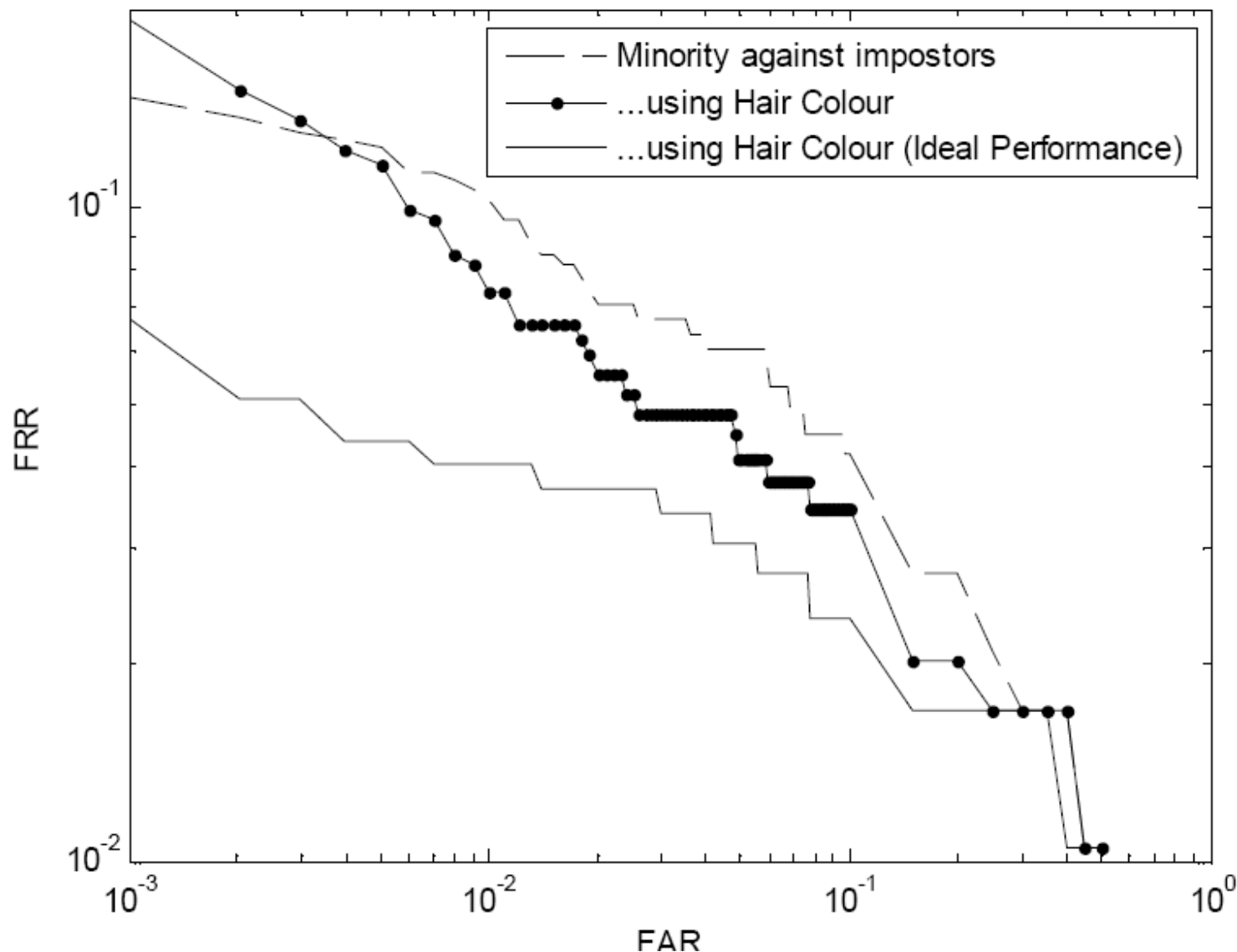
Example



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Using the previous model...

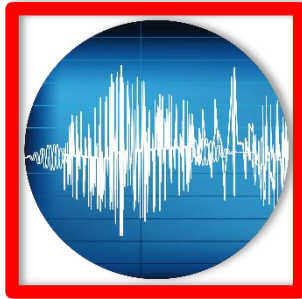


Homework

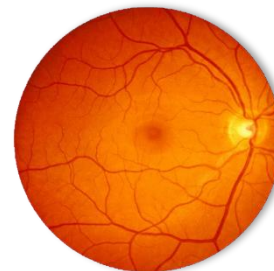


- Download some face recognition algorithms
- Download some hair colour detector, such as:
 - <https://github.com/asimadnan/Hair-Colour-Detection-with-Deep-Learning>
- You may also try some eye colour detector, such as:
 - <https://github.com/ghimiredhikura/Eye-Color-Detection>
- Try to adapt this information when looking at minority groups as previously described, using publicly available data sets

Behavioural vs. physiological biometrics



*Something you
«behave»*



Behavioural biometrics

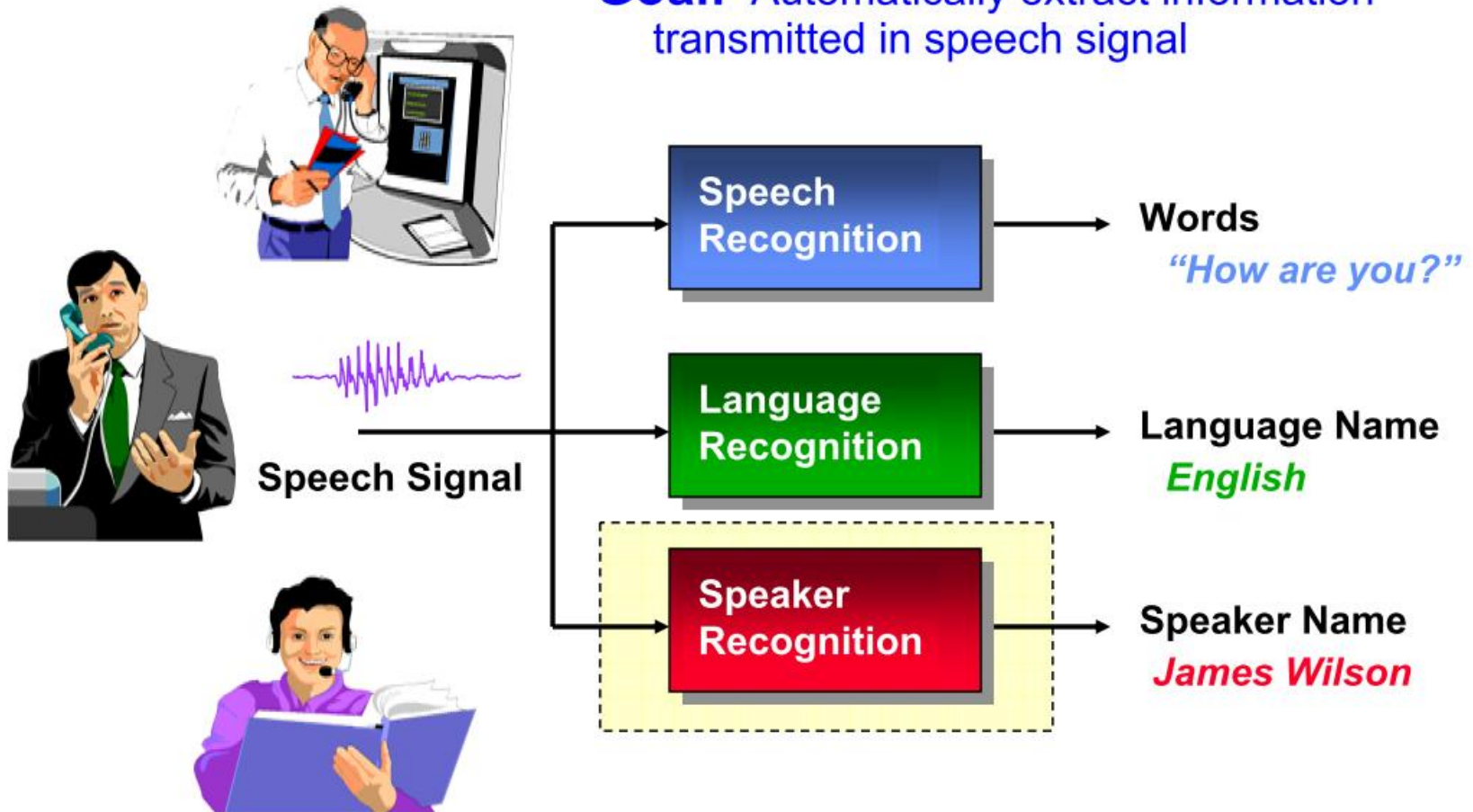


	Universality	Uniqueness	Permanence	Collectability	Performance	Acceptability	Circumvention	Cost	Maturity
Fingerprint	M	H	H	M	H	M	L	M	H
Face	H	L	M	H	L	H	H	L	M
Iris	H	H	H	M	H	L	L	H	M
Hand vein	M	M	M	M	M	M	H	H	L
Facial thermogram	H	H	L	H	M	H	H	H	M
Retinal scan	H	H	M	L	H	L	H	H	H
Ear	M	M	H	M	M	H	M	M	L
Hand geometry	M	M	M	H	M	M	M	L	H
Voice	M	L	L	M	L	H	H	L	H
Signature	L	L	L	H	L	H	L	M	M
Gait	M	L	L	H	L	H	M	H	L

Voice recognition

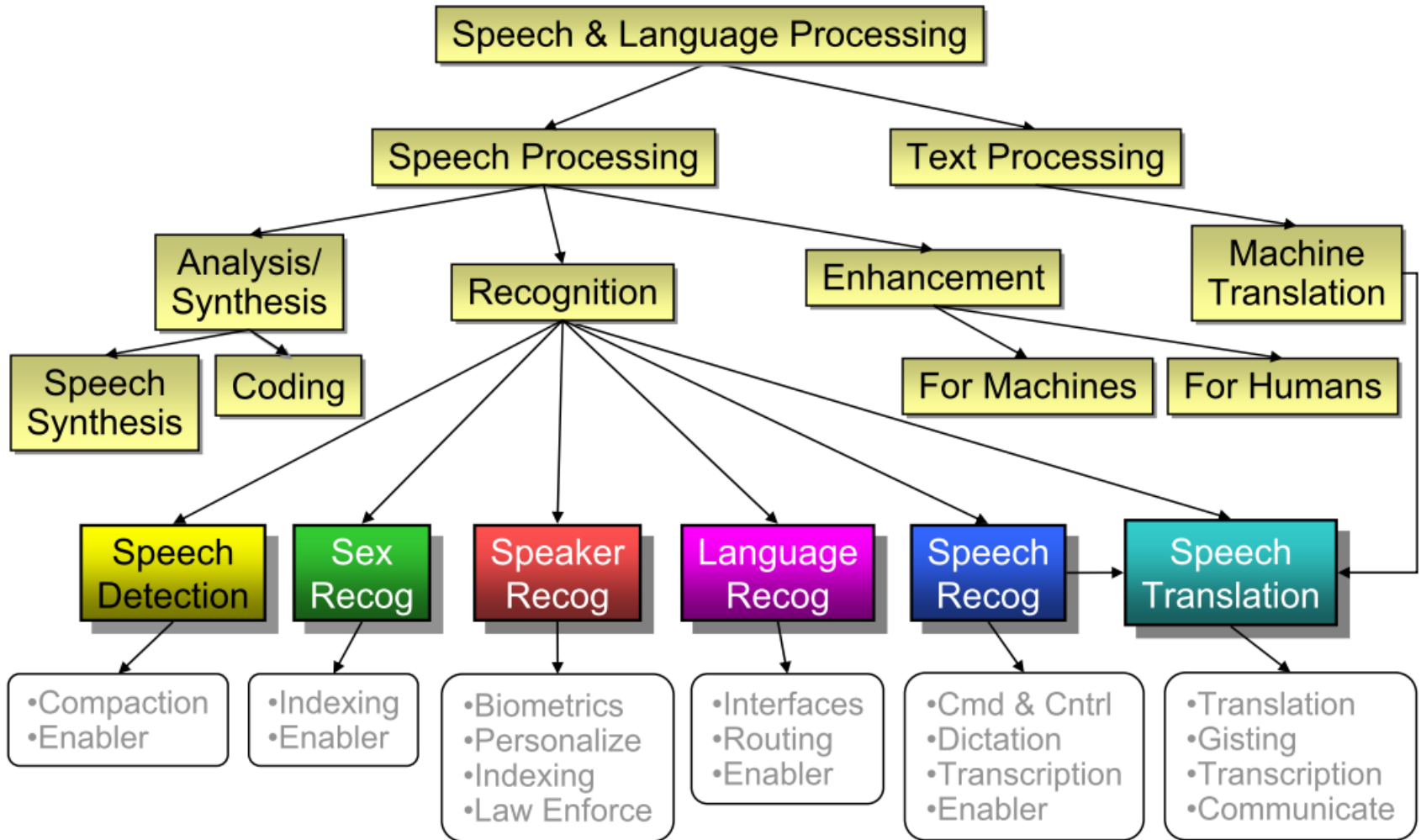


Goal: Automatically extract information transmitted in speech signal



From the lecture by D. Reynolds at
JHU Summer School 2010

Taxonomy



Hypothesis

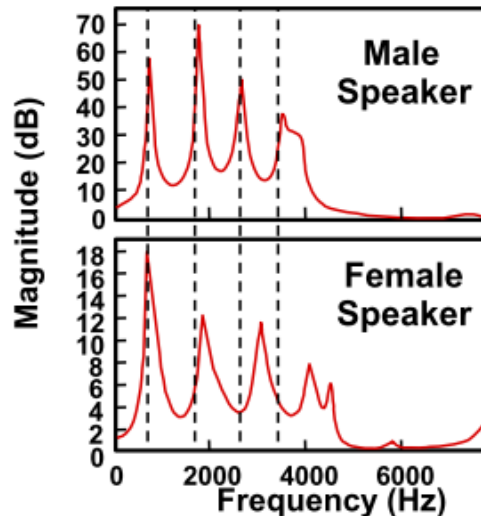


- Different frequency profiles when speaking the same word depending on the gender, the age and the people's identity

Cross Section of
Vocal Tract



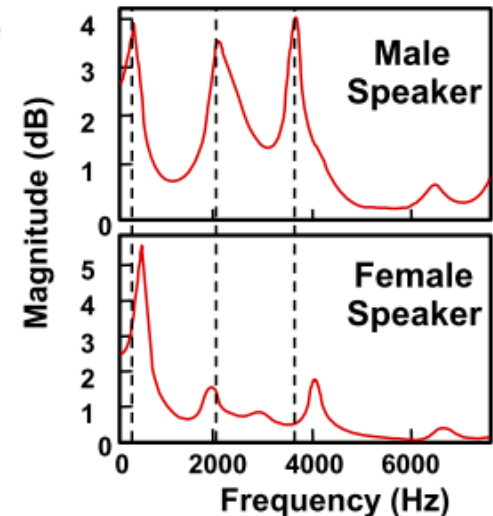
/AE/



Cross Section of
Vocal Tract



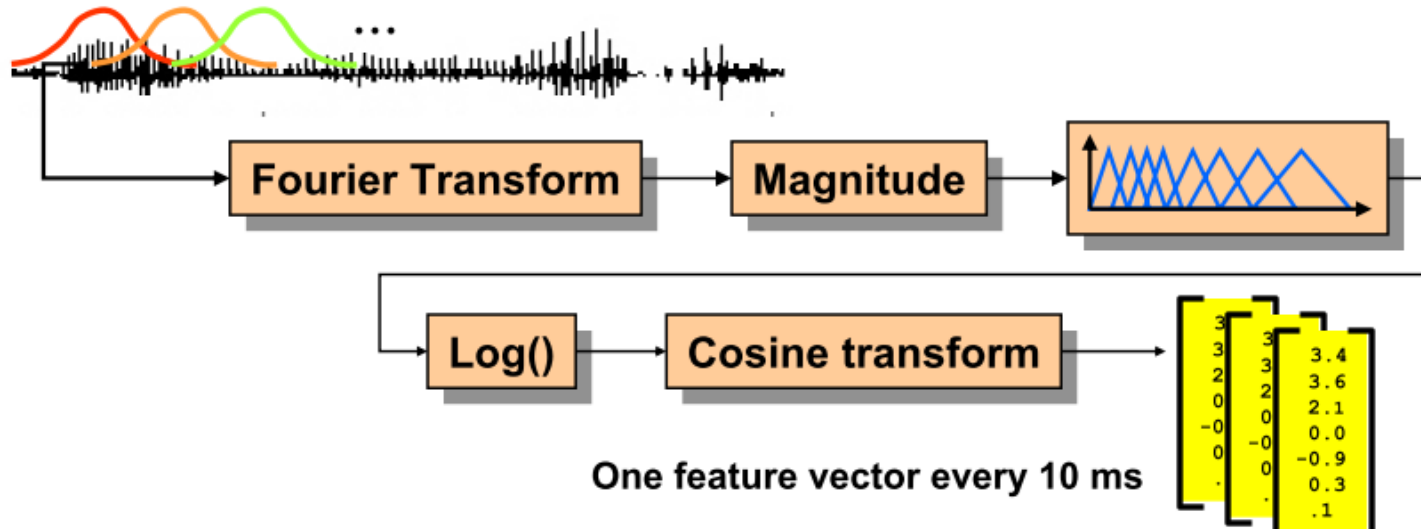
/I/



Approaches to feature extraction and classification



- **Spectral-based**



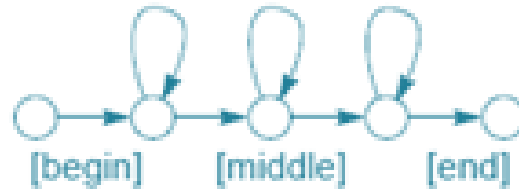
- **Classification**

- Hidden Markov Models (HMM)
- Support Vector Machines (SVM)

Hidden Markov Models



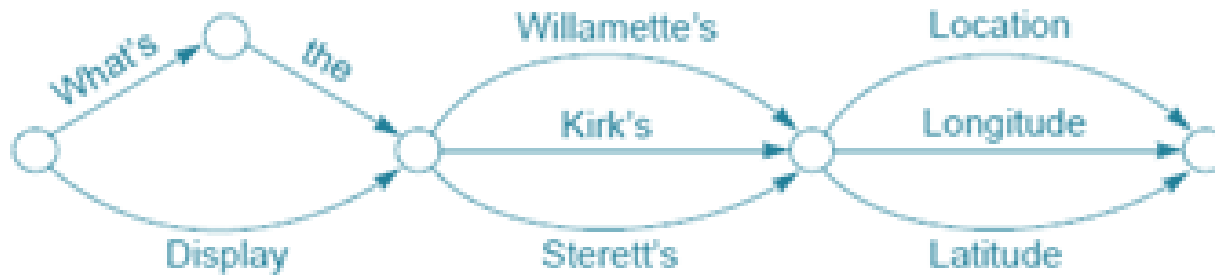
Phoneme
level



Word
level



Sentence
level



PA and attacks against voice

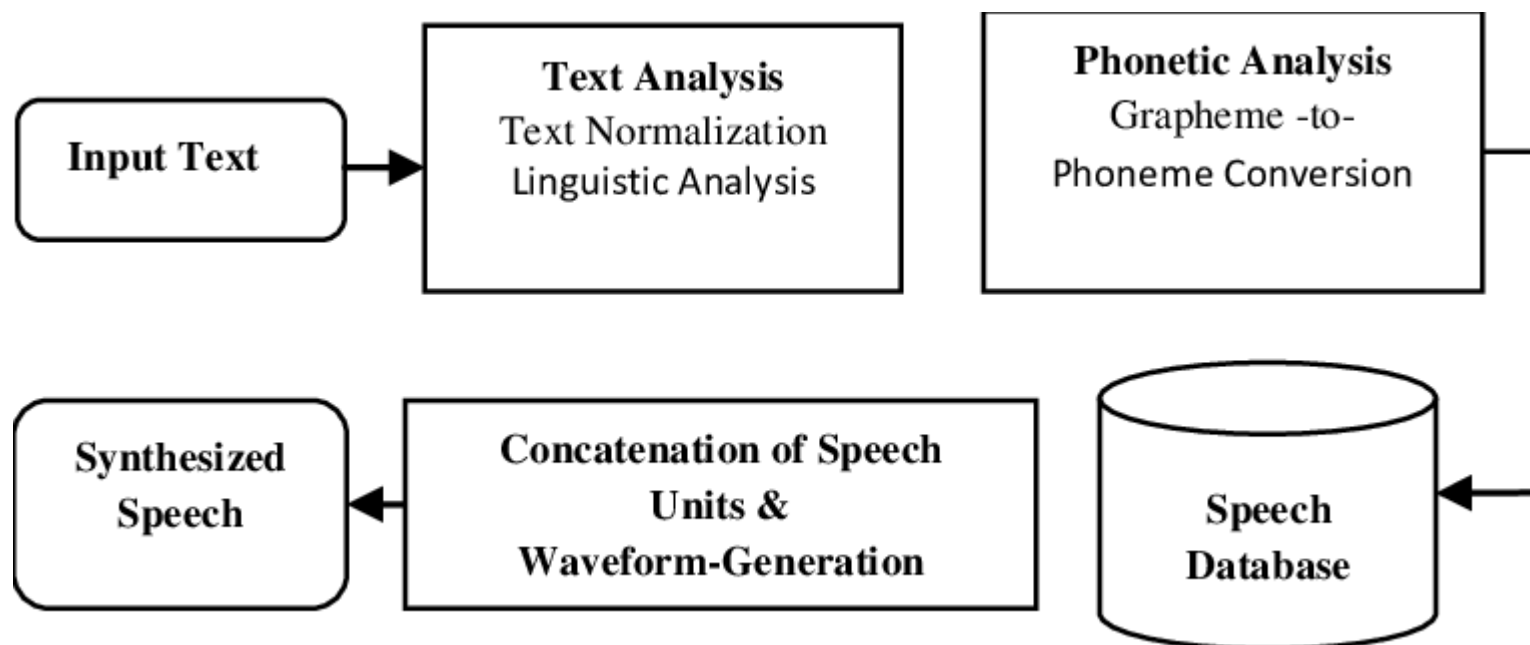


- **Presentation attacks**

- Replay attack
- Imitator
- Synthesis: Text-To-Speech and Voice Conversion

- **Poisoning**

- **Adversarial attacks**

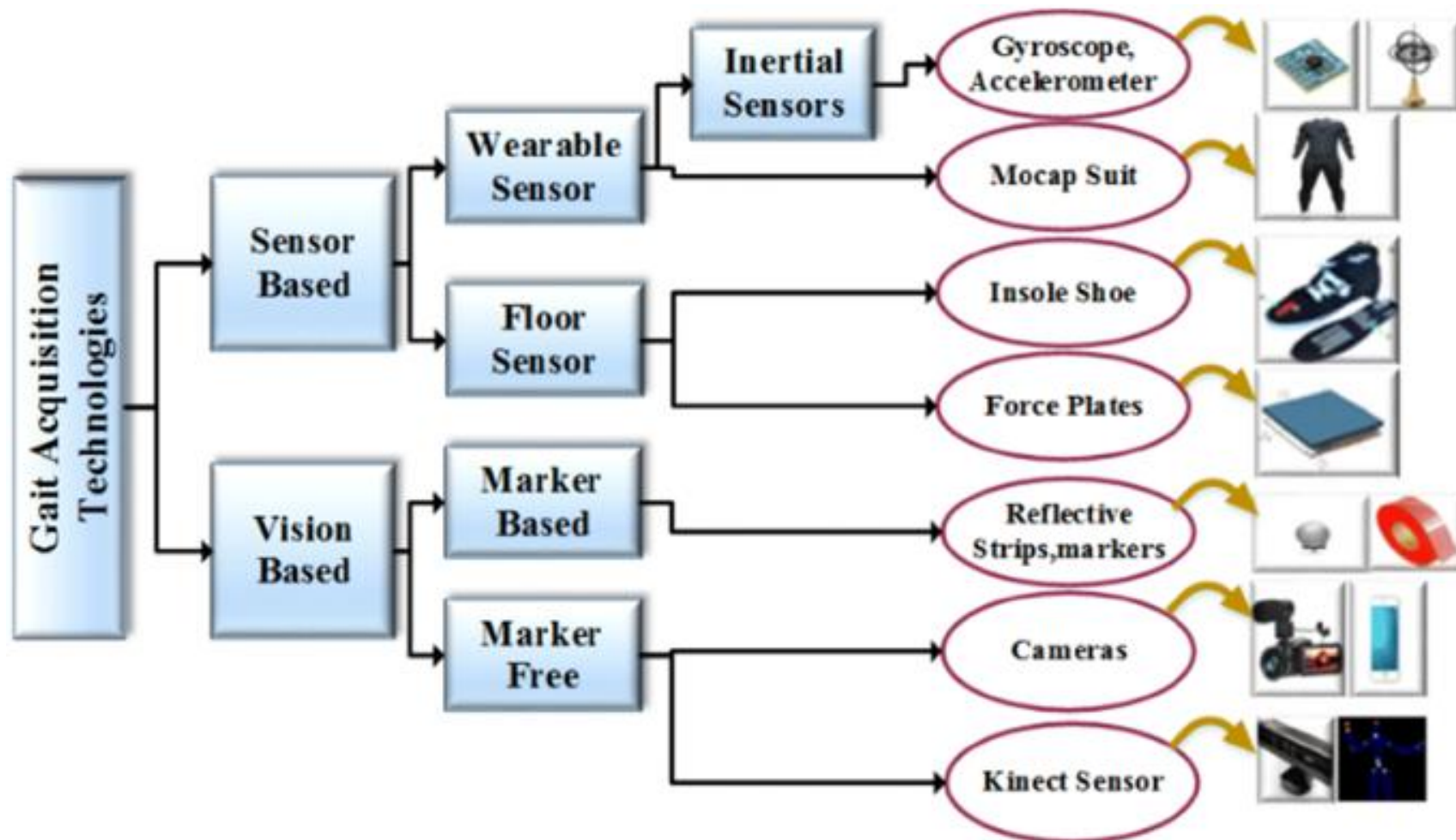


Homework

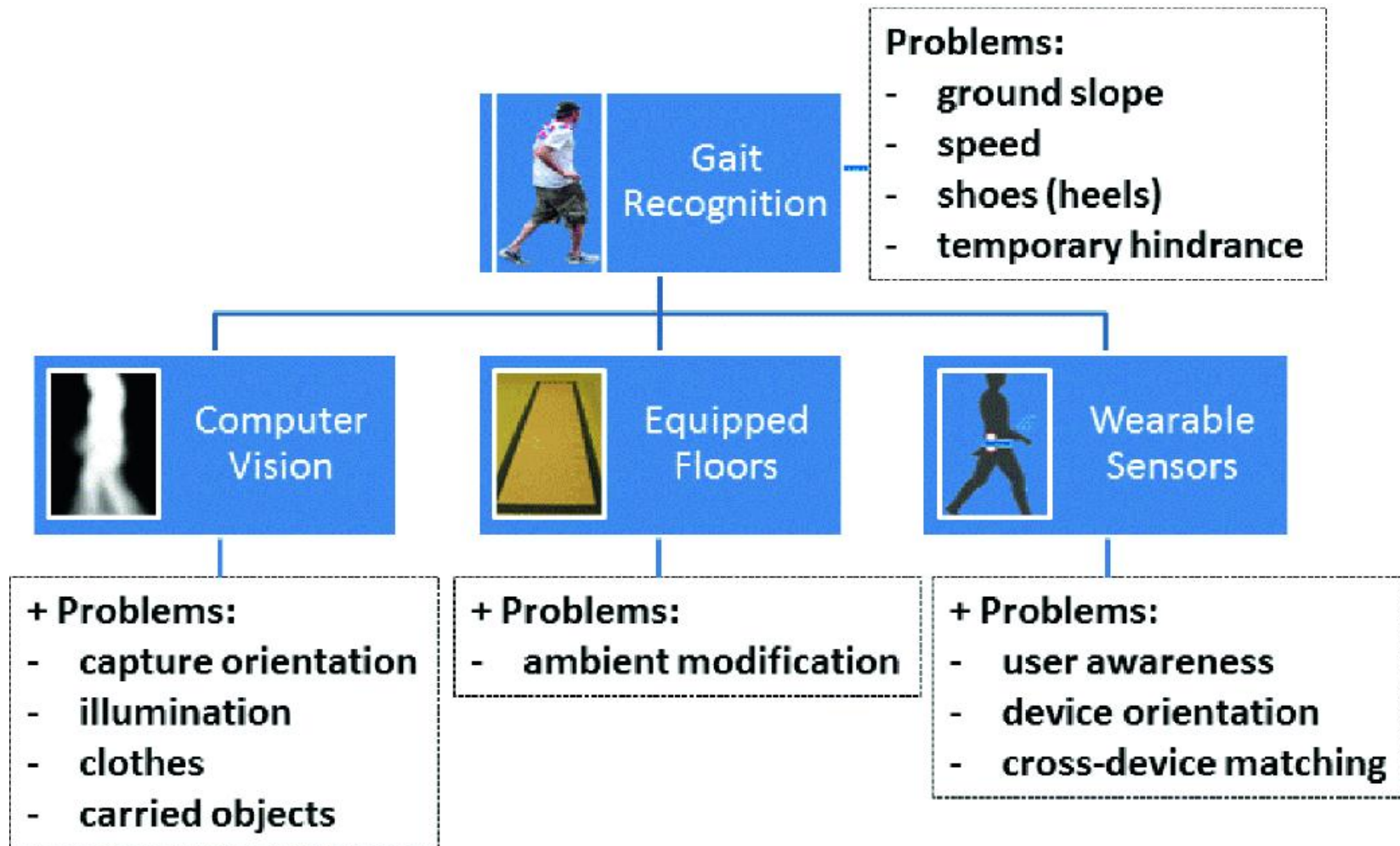


- Download some TTS and VC systems available on the web
- Download SR algorithms and train it on your voice
- Try to generate your own voice synthetically using TTS and VC approaches
- Compute the match scores average and deviation when the system is attacked (TTS/VC) against the same values when it is not (your voice)

Gait recognition

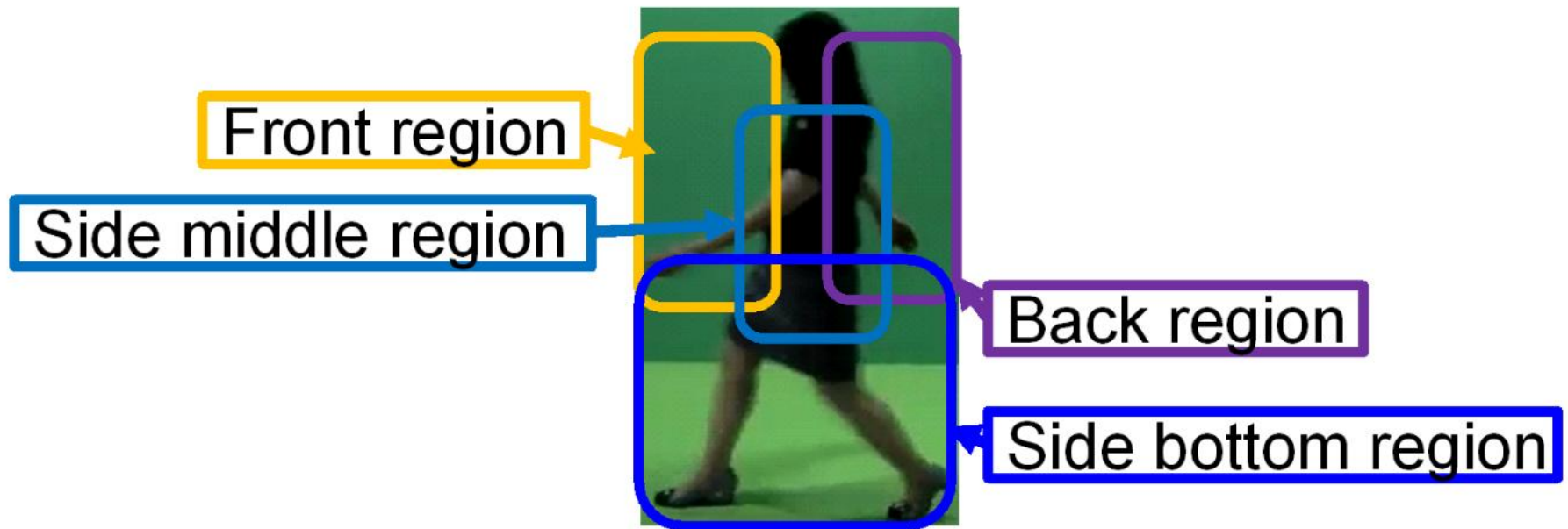


Problems





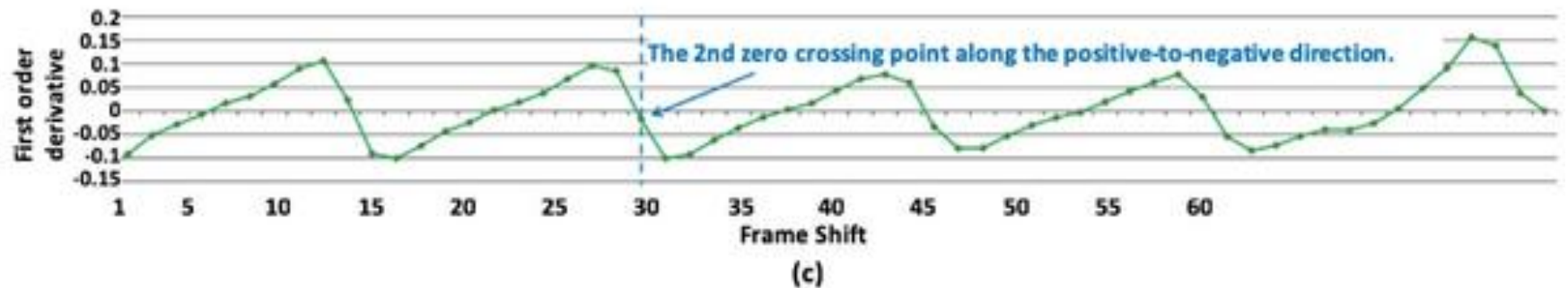
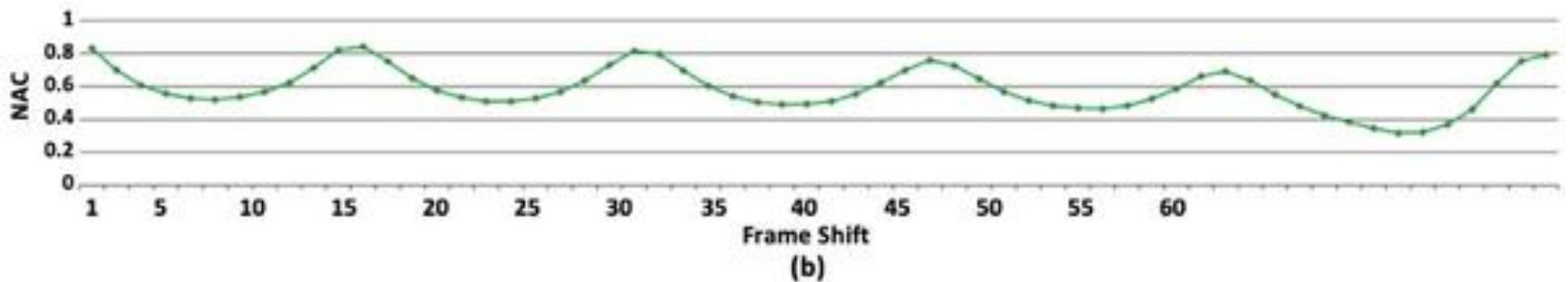
- Usual acquisition by «profile» view
- This view allows pointing out presence of:
 - Clothes
 - Objects



Rationale behind profile-based acquisition



(a)



Gait Energy Image



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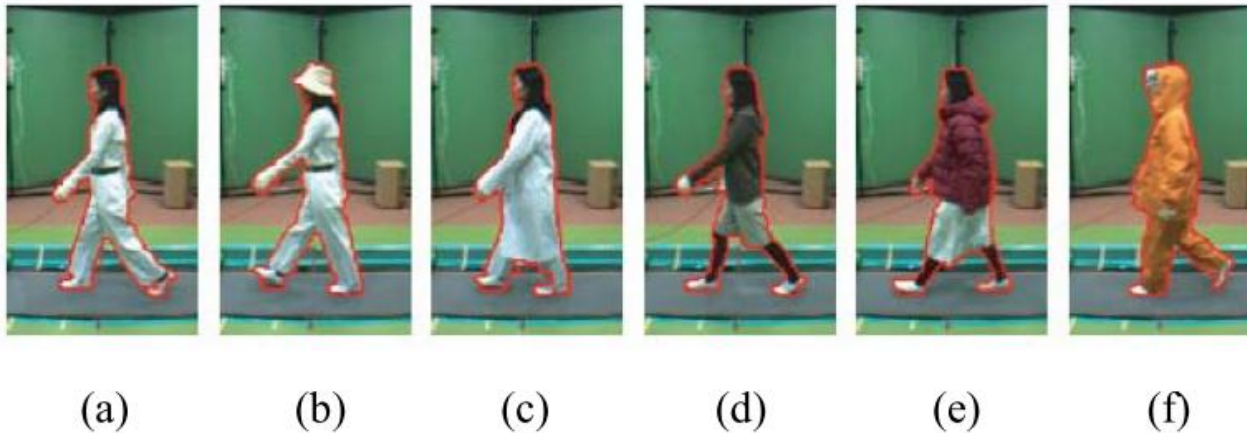


Figure 1. Sample images of clothing from OU-ISIR Treadmill dataset B of a subject [7].

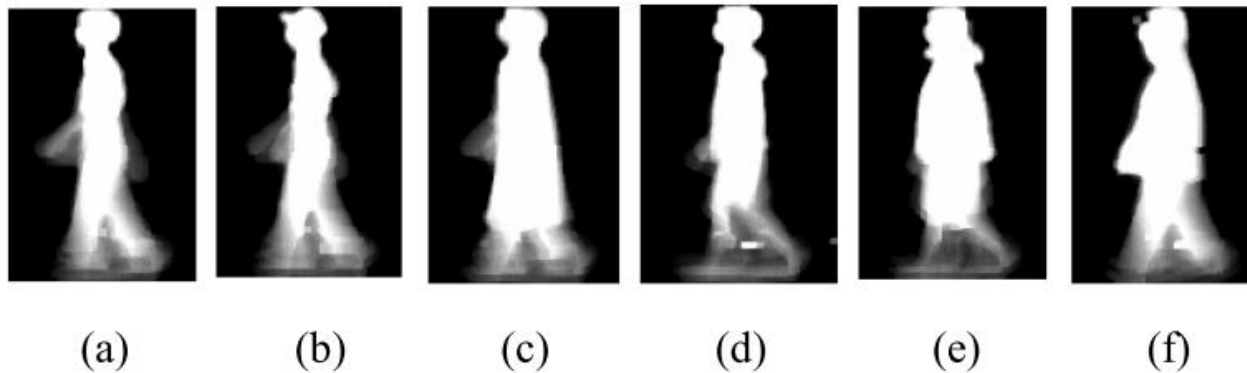


Figure 2. The Gait Energy Image version of Fig. 1.

- Given the silhouette image $B(t)$ at time t , and being T the period associate to a complete gait cycle, Gait Energy image is computed by the following formula:

$$G = \frac{255}{T} \sum_{t=1}^T B(t)$$

- Feature set G can be a first-level «template» for each people
- However, usually further processing steps are done to refine features embedded in each GEI

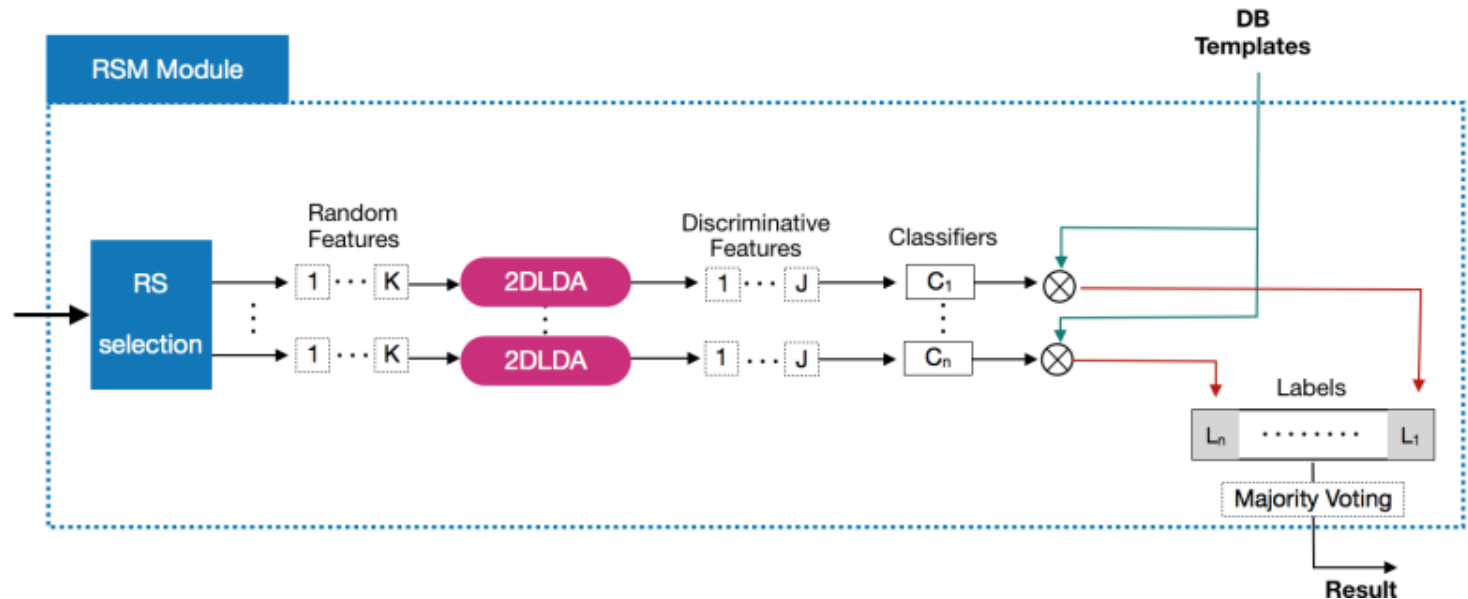
Multi-modal gait recognition



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GEI

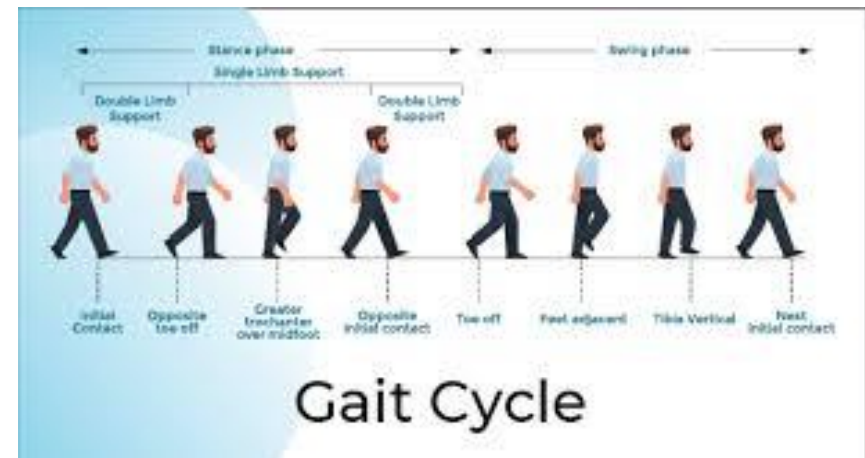


Method	Normal Conditions	Carrying Conditions	Clothing Conditions	Overall	Std
Yu et al. [17]	97.60	32.70	52.00	60.77	33.33
Han et al. [16]	99.60	57.20	23.80	60.20	37.99
Bashir et al. [21]	100.00	78.30	44.00	74.10	28.24
Bashir et al. [22]	97.50	83.60	48.80	76.63	25.09
Bashir et al. [23]	99.40	79.90	31.30	70.20	35.07
Dupuis et al. [24]	98.80	73.80	63.70	78.77	18.07
Rida et al. [32], [33]	93.60	81.70	68.80	81.37	12.40
Rida et al. [25]	95.97	63.39	72.77	77.38	16.77
Hu et al. [12]	94.00	45.20	42.90	60.70	28.86
Kusakunniran [27]	95.40	60.90	52.00	69.43	22.92
Rakanujjaman et al. [29]	97.61	83.87	51.61	77.70	23.61
Kusakunniran [28]	94.50	60.90	58.50	71.30	20.13
Jeevan et al. [26]	93.36	56.12	22.44	57.31	35.47
Our Method	98.39	75.89	91.96	88.75	11.59

Group Homework



- Capture your gait by a webcam :
profile view is ok
 - Build a small data set of yours □
- Segment frames and compute the GEI...
- Replicate some of previously shown results...
- OR design your own embeddings
by handcrafting or fine tuning a pre-trained network
- Try also to deceive the system for
obtaining lower match scores

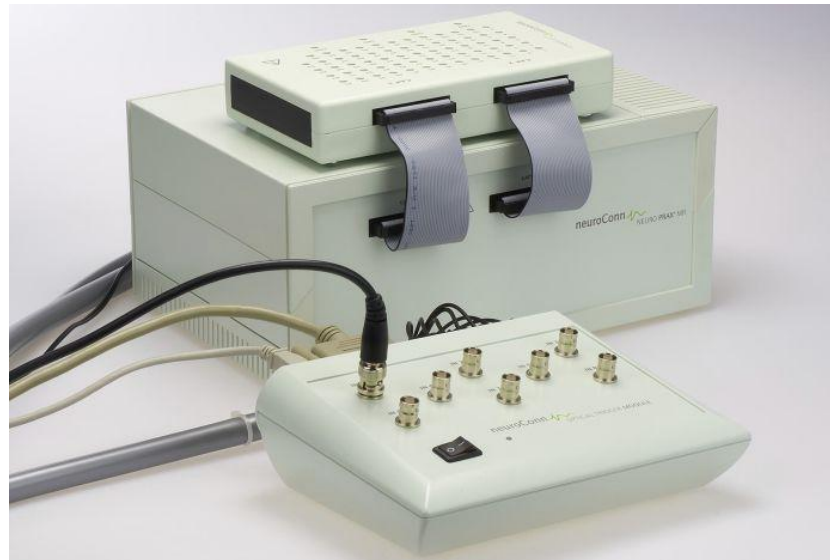
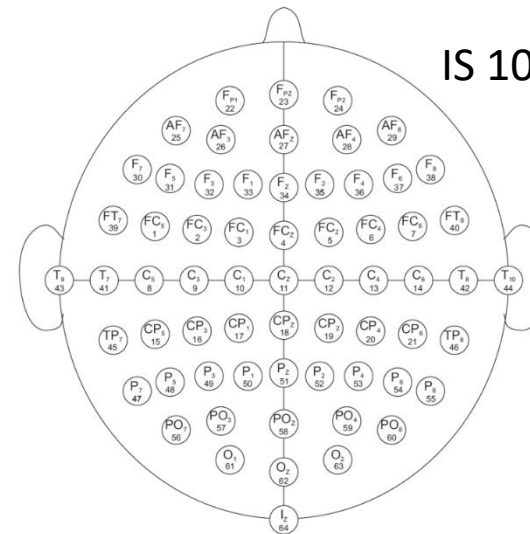


- **EEG is the record of the electrical activity of the brain from the scalp**
 - Cortical electrical activity
- **Disorders diagnosis and treatment**
 - Brain tumor
 - Brain damage from head injury
 - Brain dysfunction that can have a variety of causes (encephalopathy)
 - Inflammation of the brain (encephalitis)
 - Stroke (loss of consciousness → thrombosis)
 - Sleep disorders
 - Amyotrophic Lateral Sclerosis (SLA)
 - Also proposed for epilepsy and other brain disorders
- **Personal recognition**

Standard capture device



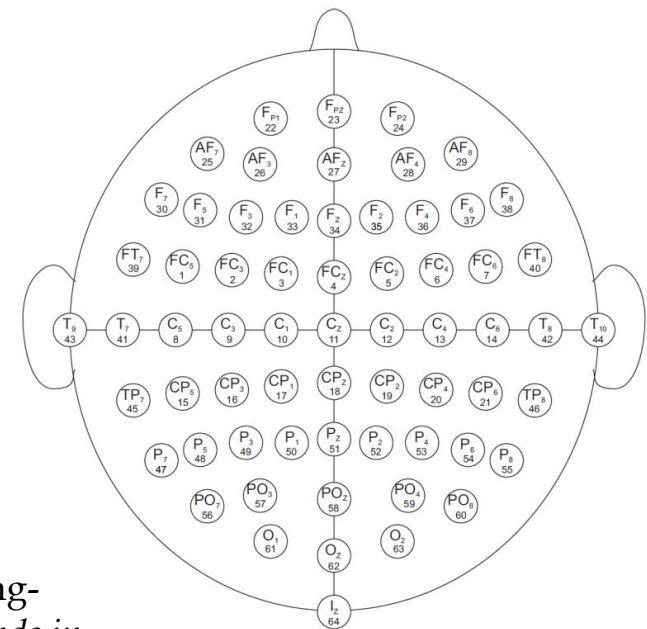
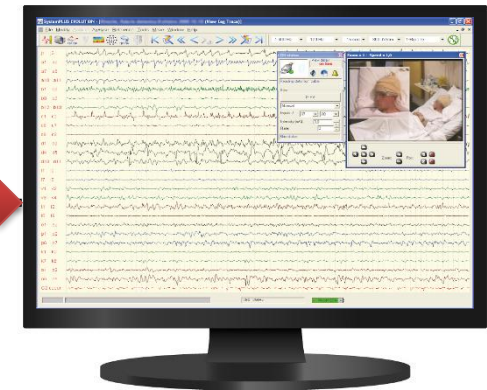
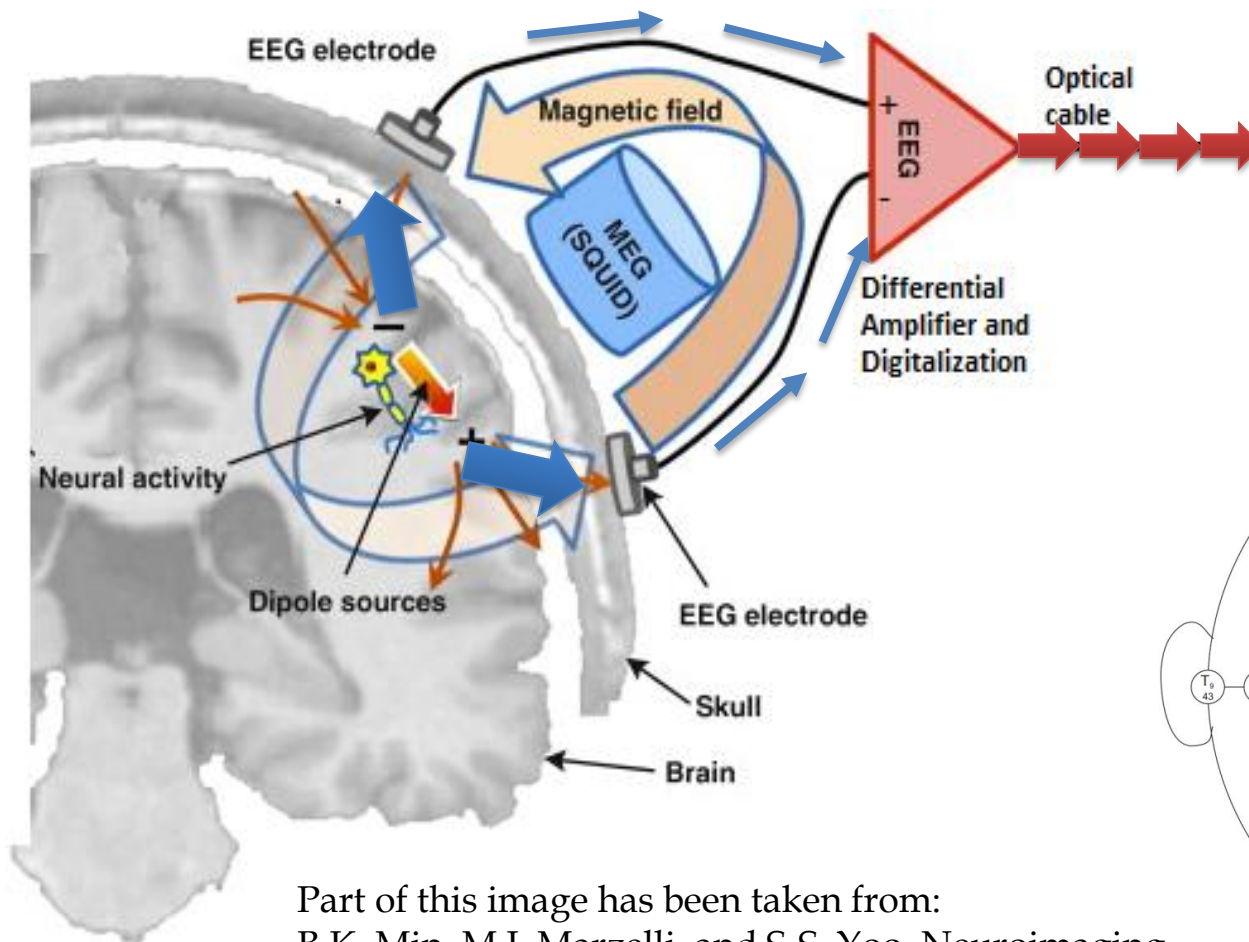
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EEG sourcing



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Part of this image has been taken from:
B.K. Min, M.J. Marzelli, and S-S. Yoo, Neuroimaging-
based approaches in brain-computer interface, *Trends in
Biotechnology*, 28(11):552-60, 2010.

EEG for personal recognition



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- It has been shown that EEG retains significant personal information
- «Portable» capture devices:



Emotiv



NeuroSky



Zeo



Starlab



EmSense



nia Game Controller



Mindo 4



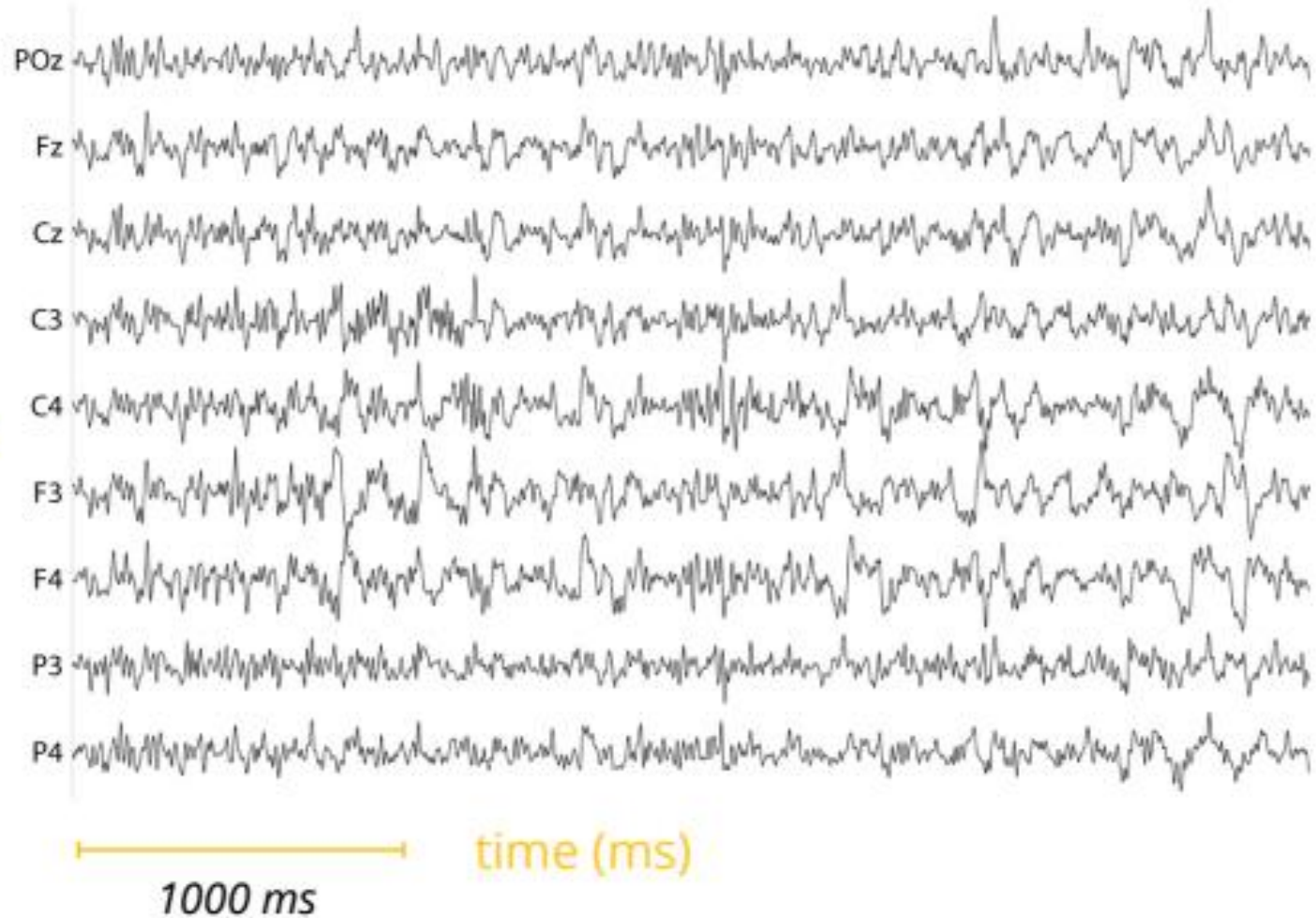
Mindo 16

EEG signal characteristics



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electrodes



EEG signal characteristics

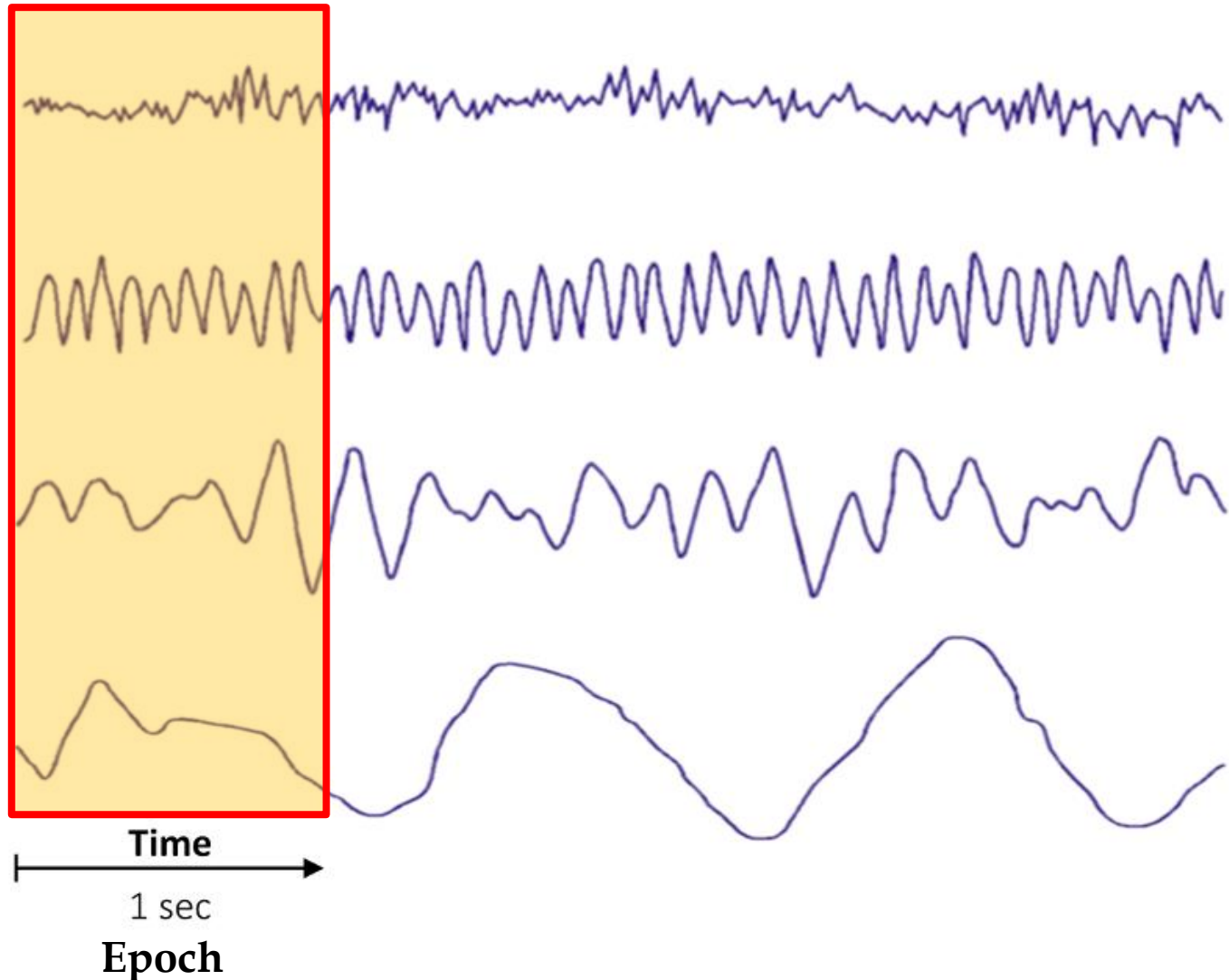


Beta
[12-30 Hz]

Alpha
[8-12 Hz]

Theta
[4-8 Hz]

Delta
[1-4 Hz]

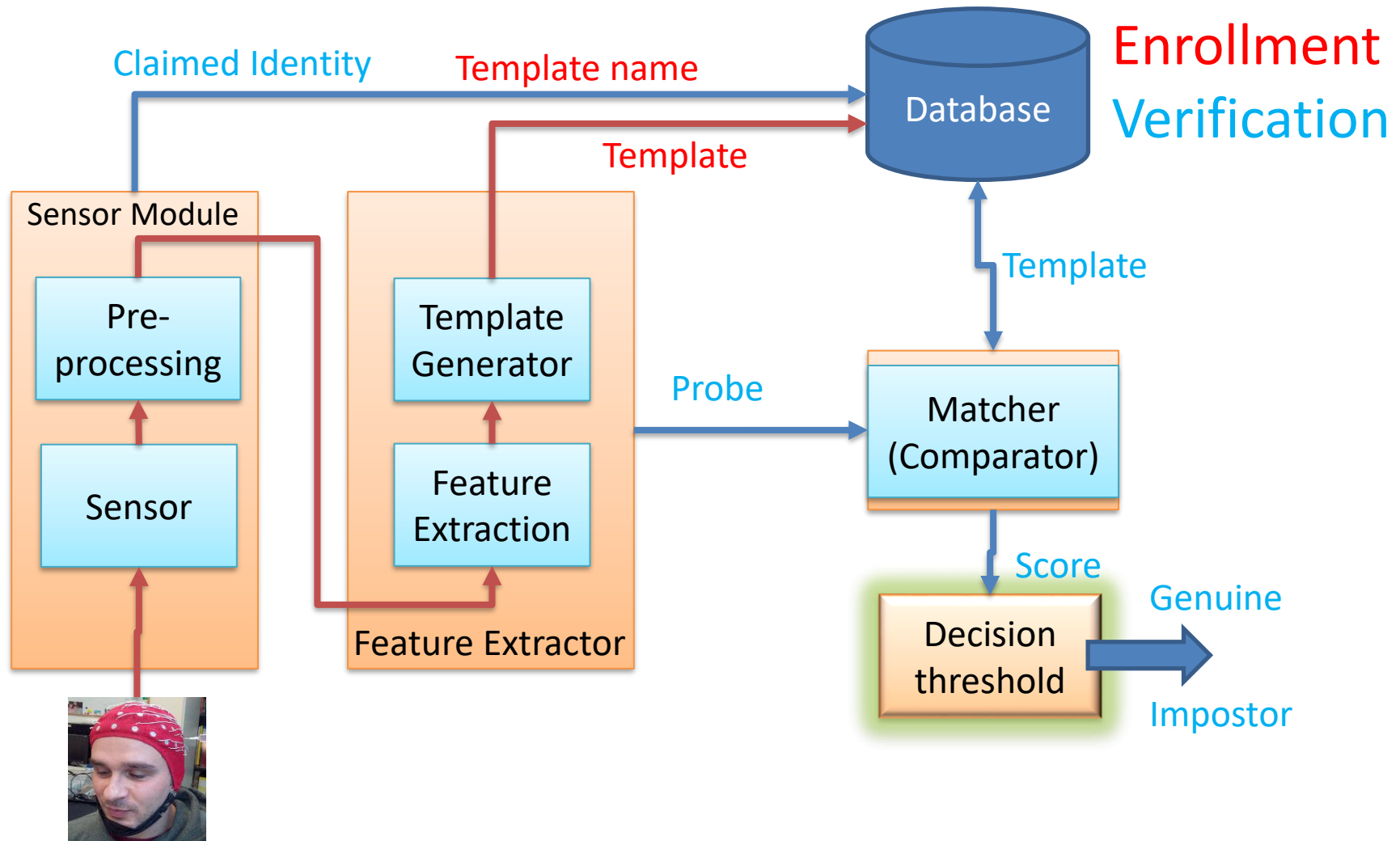




- **«Early» works (2010-2013)***
 - Same features used for biomedical applications
 - Auto-regressive coefficients
 - Power Spectrum measurements
 - Discrete Cosine Transform
 - Independent Component Analysis
- **Recent works (2014-2022)**
 - The brain as a complex network: connectivity measurements
 - Spectral coherence and pooled covariance matrix (2014)
 - Phase Lag Index and eigenvector centrality (2015)
 - Deep learning (from 2018)

*P. Campisi, and D. La Rocca, «Brain Waves for Automatic Biometric-Based User Recognition», *IEEE Trans. On Information Forensics and Security*, 9 (5) 782-800, 2014.

An EEG-based personal verification system





- **Denoising**
 - Artifacts due to movements of eye (blink) or other muscular movements
 - ICA (Independent Component Analysis)
- **Downsampling**
 - Normalizing the number of samples per second
- **Subbands/Broadband**
 - In order to focus on the most informing band for the task
 - Alfa, beta, gamma, delta

Features extraction: basics



- **Power Spectral Density**
 - over the alfa-gamma subbands

- **Wavelet transformations**

$$C_{a,b} = \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2^{-j}}} \right) \psi \left(\frac{t - 2^{-j}k}{2^{-j}} \right)$$

- Wavelet Energy

$$p_j = \frac{\sum_k^N |C_j(k)|^2}{\sum_j \sum_k |C_j(k)|^2}$$

- Wavelet Entropy

$$S_{wt}(p|q) = \sum_j p_j \cdot \ln \left(\frac{p_j}{q_j} \right)$$

Features extraction: connectivity measurements



- **Correlation coefficient among channels**

- **Phase Lag-Index**

- quantify the asymmetry of the distribution of phase differences between two signals and removes the effect of amplitude information

$$PLI(i, j) = |\langle \text{sign}[\Delta\phi_{(i,j)}(t_k)] \rangle|$$

- **Amplitude Envelop Correlation (Hilber transform)**

- obtained by measuring the magnitude of the analytic signal and successively the Pearson's correlation

- **Phase Locking Value**

$$PLV(i, j) = \left| \frac{1}{T} \sum e^{j[\Delta\phi_{(i,j)}(t_k)]} \right|$$

Spectral coherence connectivity



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EEG

Preprocessing

Feature extraction

Matching

Downsampling
filter

Coherences
Computation

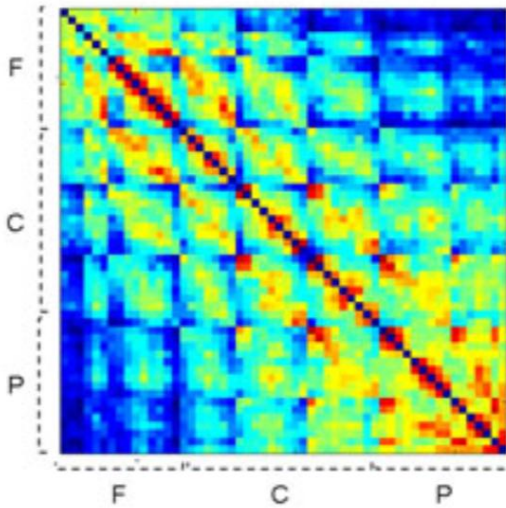
Z Fisher
normalization

Pooled
covariance
matrix

Mean feature
vector
distribution

Mahalanobis
distance

Match Score



D. La Rocca, and P. Campisi. «Human Brain Distinctiveness Based on EEG Spectral Coherence Connectivity». *IEEE Trans. on Biomedical Engineering*, 61 (9) 2014.

Eigenvector Centrality

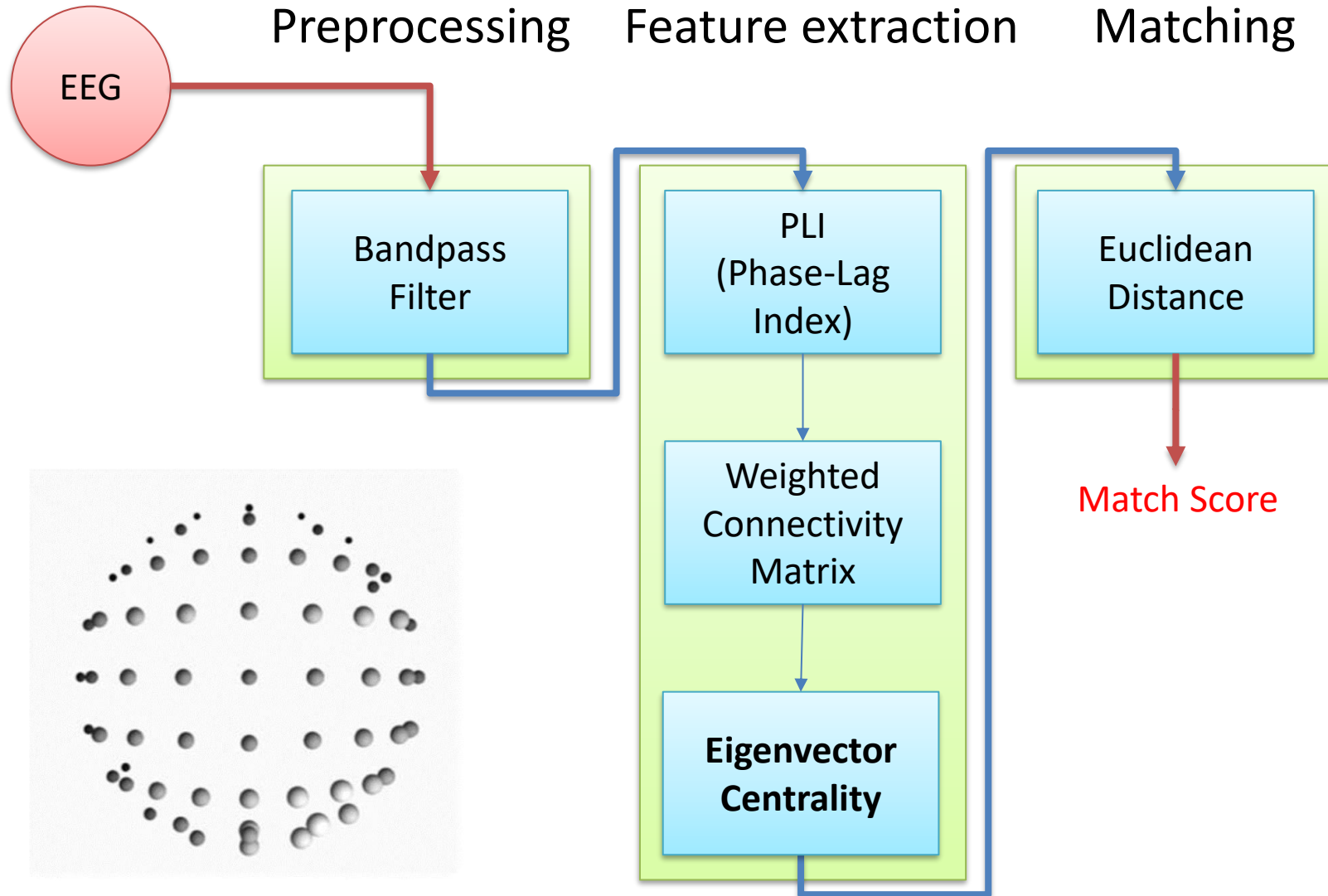


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Preprocessing

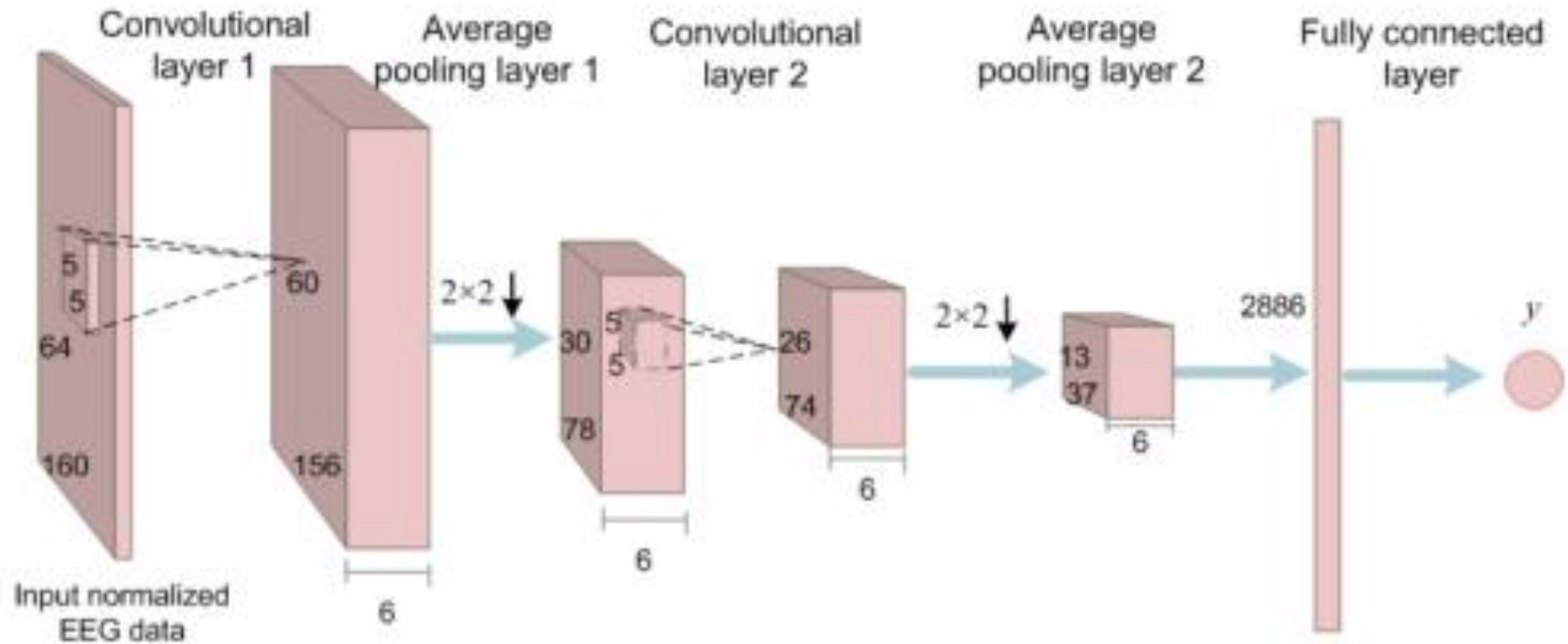
Feature extraction

Matching



M. Fraschini, A. Hillebrand, M. Demuru, L. Didaci, and G.L. Marcialis. «An EEG-Based Biometric System Using Eigenvector Centrality in Resting State Brain Networks». *IEEE Signal Processing Letters* 22 (6): 666–70, 2015.

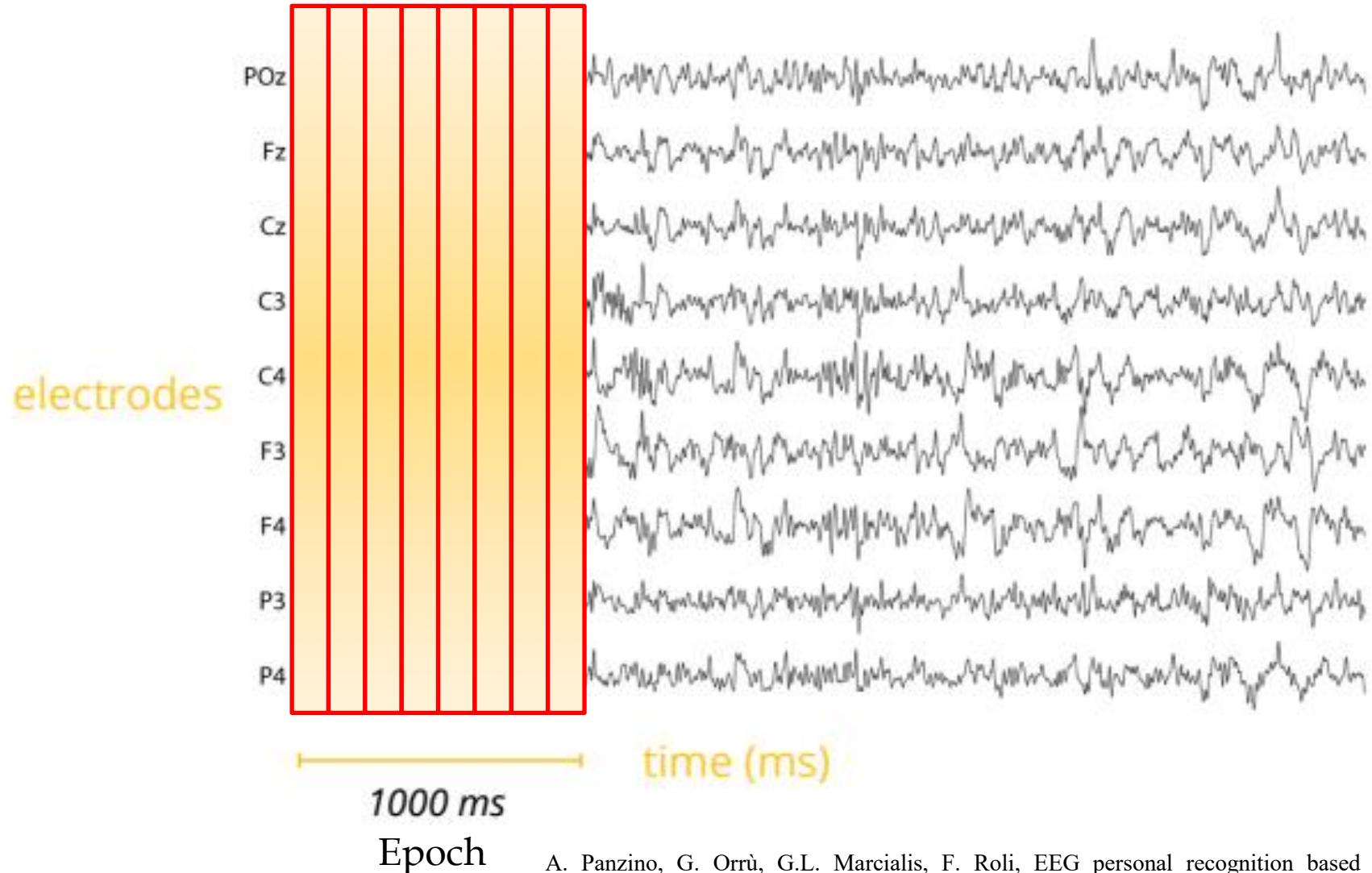
EEG personal recognition based on CNN



Patch-based EEG recognition



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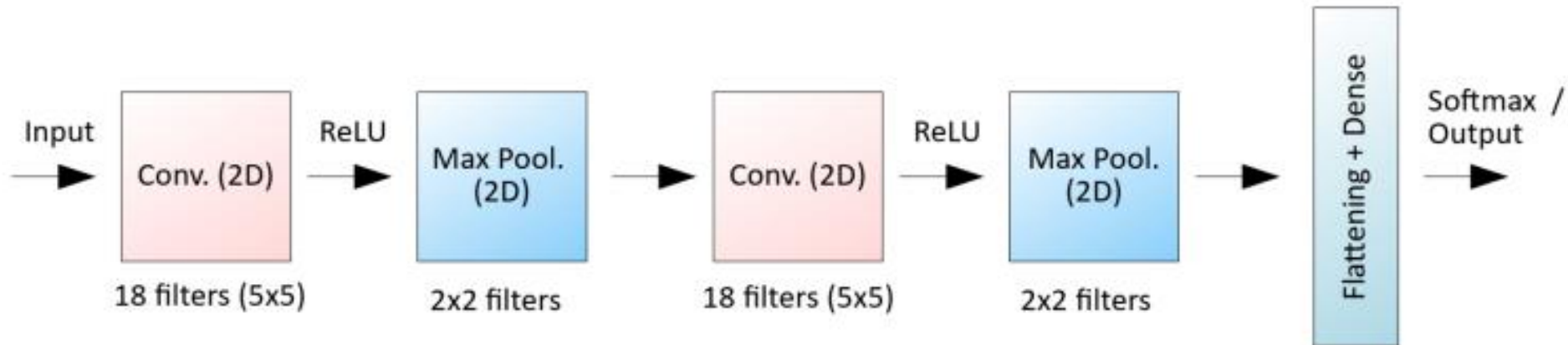
A. Panzino, G. Orrù, G.L. Marcialis, F. Roli, EEG personal recognition based on 'qualified majority' over signal patches, *IET Biometrics*, IET, open access at: <https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/bme2.12050>, 2021

CNN and patch-based personal recognition from EEG



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- A possible architecture developed in our lab

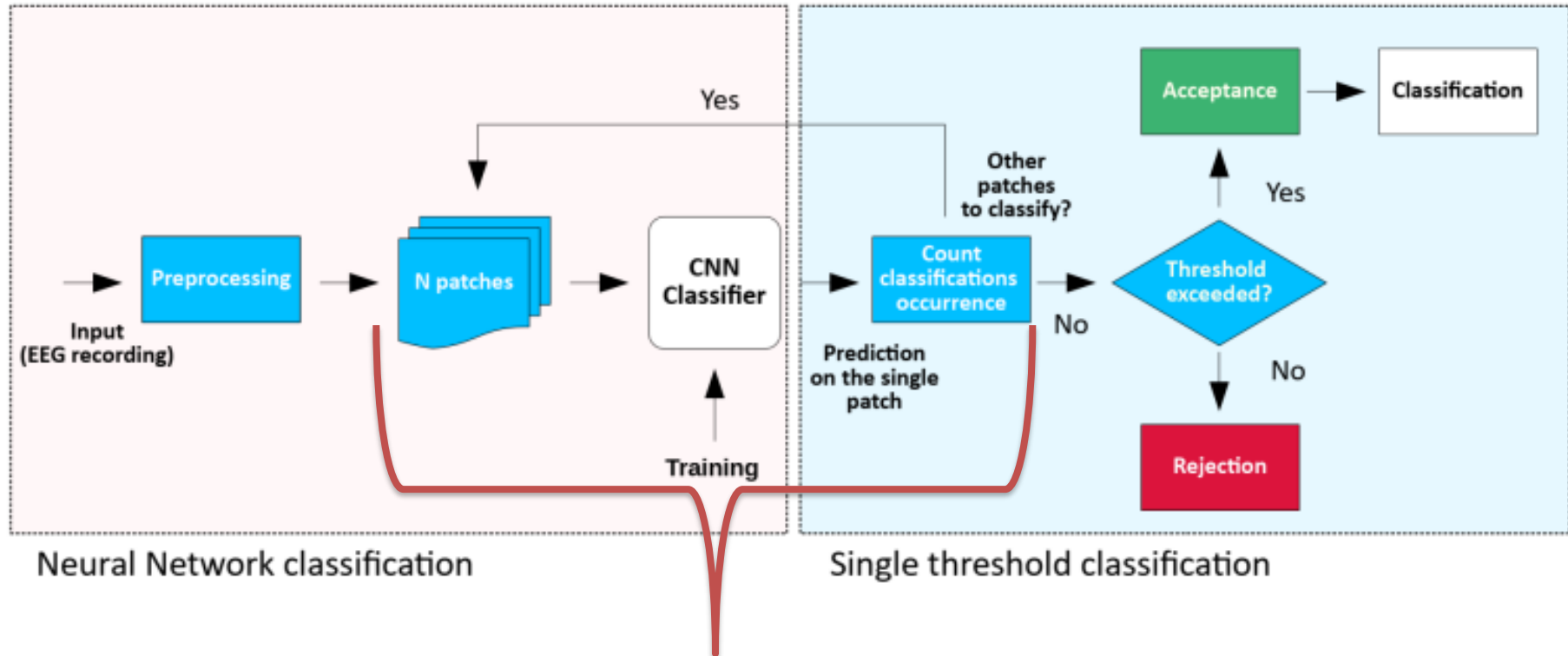


Total params: 40,321

Trainable params: 40,321

Non-trainable params: 0

Decision-level fusion



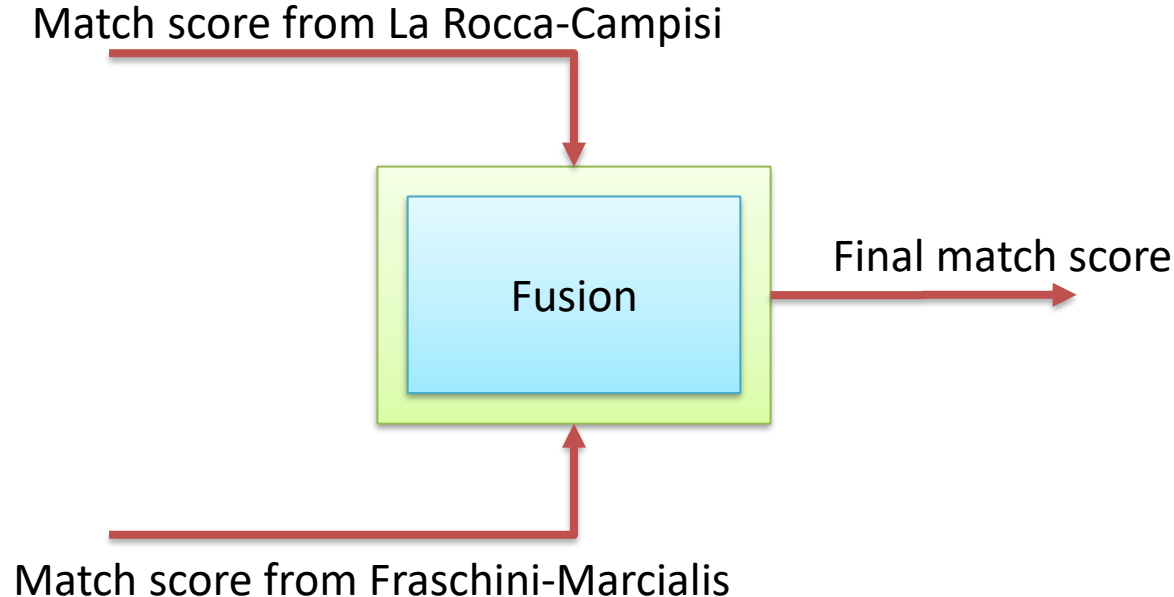
Pipeline

- How much time does take the input recording?
- How much time does take the patch classification?
- Where is the bottleneck?

Other fusion methods



- **Feature-level by concatenation**
- **Score-level by:**
 - Simple fixed rules
 - Sum, Geometric average, Max, Min, Bayes' rule
 - Likelihood Ratio rule



Data set per EEG personal recognition

EEG motor movement/imagery dataset

BCI2000¹ 64 channel Physionet². IS 10-20 system

109 subjects

14 acquisitions, 1 or 2 minutes per acquisition

Resting state at Eyes Open (EO) and Closed (EC)

Tasks 1,3 : finger movements

Tasks 2,4 : thinking to finger movements

1. Schalk, G., McFarland, D.J., Hinterberger, T., Birbaumer, N., Wolpaw, J.R. BCI2000: A General-Purpose Brain-Computer Interface (BCI) System. *IEEE Transactions on Biomedical Engineering* **51**(6):1034-1043, 2004.

2. Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PCh, Mark RG, Mietus JE, Moody GB, Peng C-K, Stanley HE. PhysioBank, PhysioToolkit, and PhysioNet; <http://circ.ahajournals.org/cgi/content/full/101/23/e215>; 2000 (June 13).

Within-task and between-task performance

Table 1

Worst (left) and best (right) within-task performance for the different connectivity metrics expressed as EER for each frequency band. EER values lower than 10% were marked as bold.

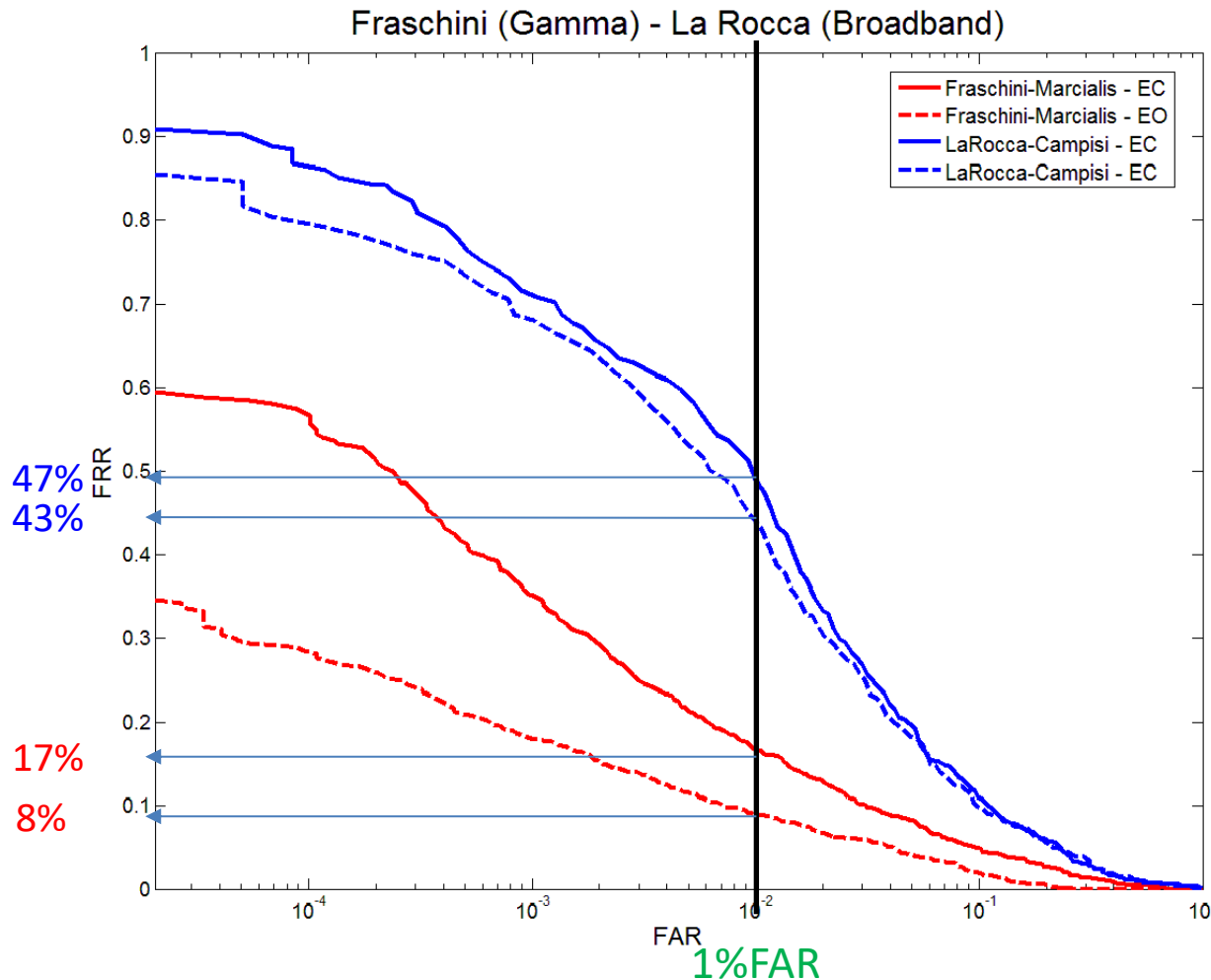
	AECc	PLI	CC	AEC	PLV
Delta	48.29–41.89	48.90–46.06	9.27–5.91	25.30–15.32	5.65–3.30
Theta	46.22–43.33	45.25–42.98	8.20–5.01	23.97–17.54	7.34–3.19
Alpha	42.79–39.38	36.58–33.43	10.17– 7.03	24.78–18.67	12.67– 5.37
Beta	36.34–26.86	14.06– 8.37	3.78–0.57	10.28– 4.94	2.37–0.09
Gamma	36.71–28.64	8.52–3.65	5.76–1.55	17.90– 7.44	2.90–0.58

Table 2

Worst (left) and best (right) between-tasks performance for the different connectivity metrics expressed as EER for each frequency band. EER values lower than 10% were marked as bold. In brackets EER values when between-tasks comparison including eyes-closed resting-state conditions were excluded from the analysis.

	AECc	PLI	CC	AEC	PLV
Delta	49.92 (49.48)–46.05	49.88 (49.82)–47.90	32.90 (28.83)–19.74	39.82 (37.22)–29.51	29.12 (25.99)–15.00
Theta	48.73–45.80	47.11 (46.63)–44.42	28.76 (26.98)–17.88	36.68–25.31	25.90 (24.19)–13.57
Alpha	47.84 (45.77)–41.67	47.92 (42.86)–37.55	41.88 (26.49)–15.67	41.04 (33.63)–23.60	40.24 (27.58)–12.56
Beta	43.59 (41.78)–35.03	34.01 (32.13)–21.89	29.96 (19.84)– 8.14	30.46 (28.24)–14.23	29.09 (18.69)– 7.38
Gamma	44.40 (44.00)–37.86	37.81 (32.27)–22.53	29.47 (24.30)–12.40	34.93 (32.64)–18.09	27.78 (22.66)–10.29

Performance difference



Elapsed time (Matlab):

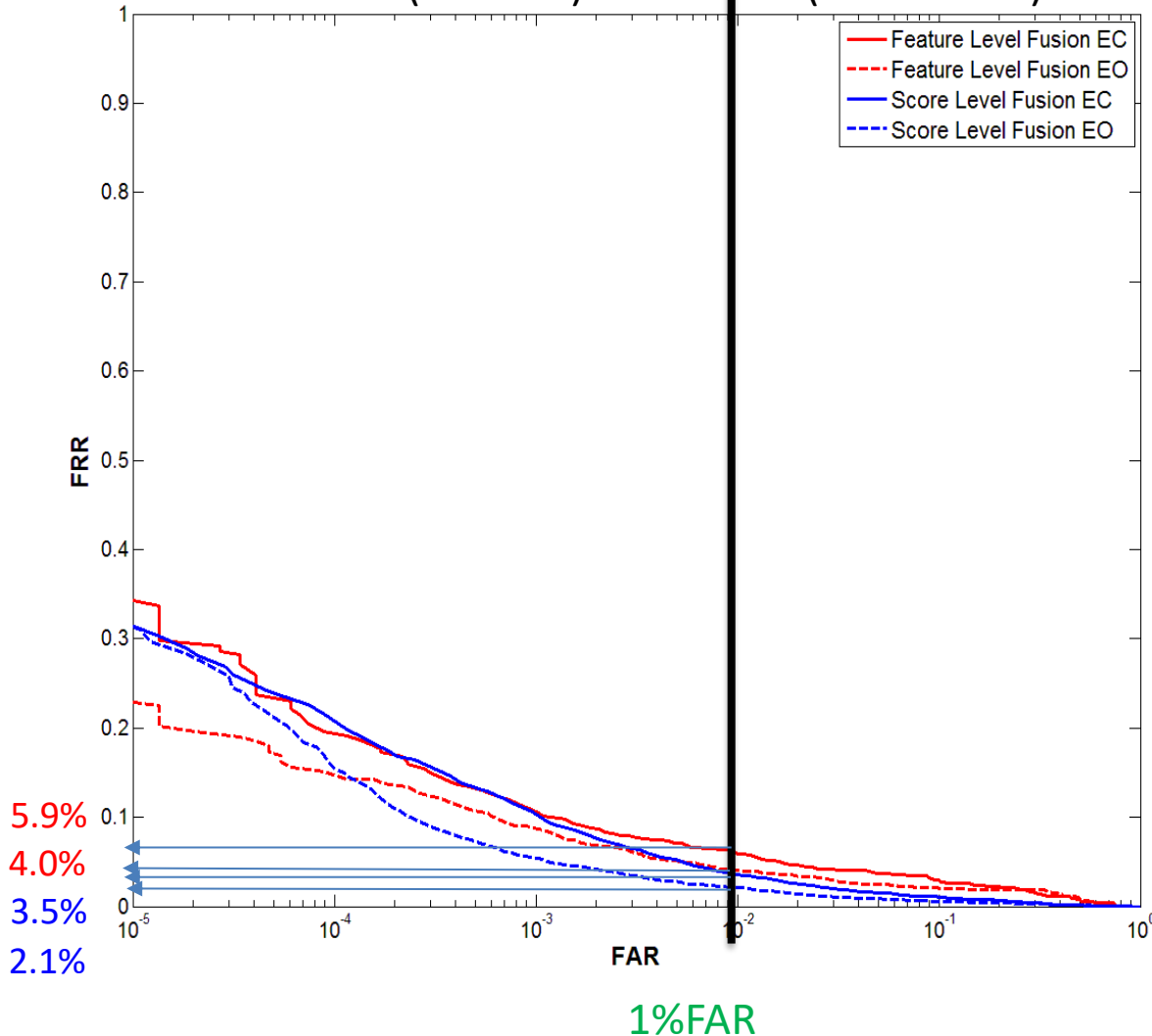
- Fraschini ~ 1 msec
- La Rocca ~ 600 msec

EER	EC	EO
Fraschini	6.51%	4.46%
La Rocca	10.6%	9.91%

Fusion: Feature level and Score level



Fraschini (Gamma) – La Rocca (Broadband)



- **Feature Level by concatenation of feature vectors (FLF):**

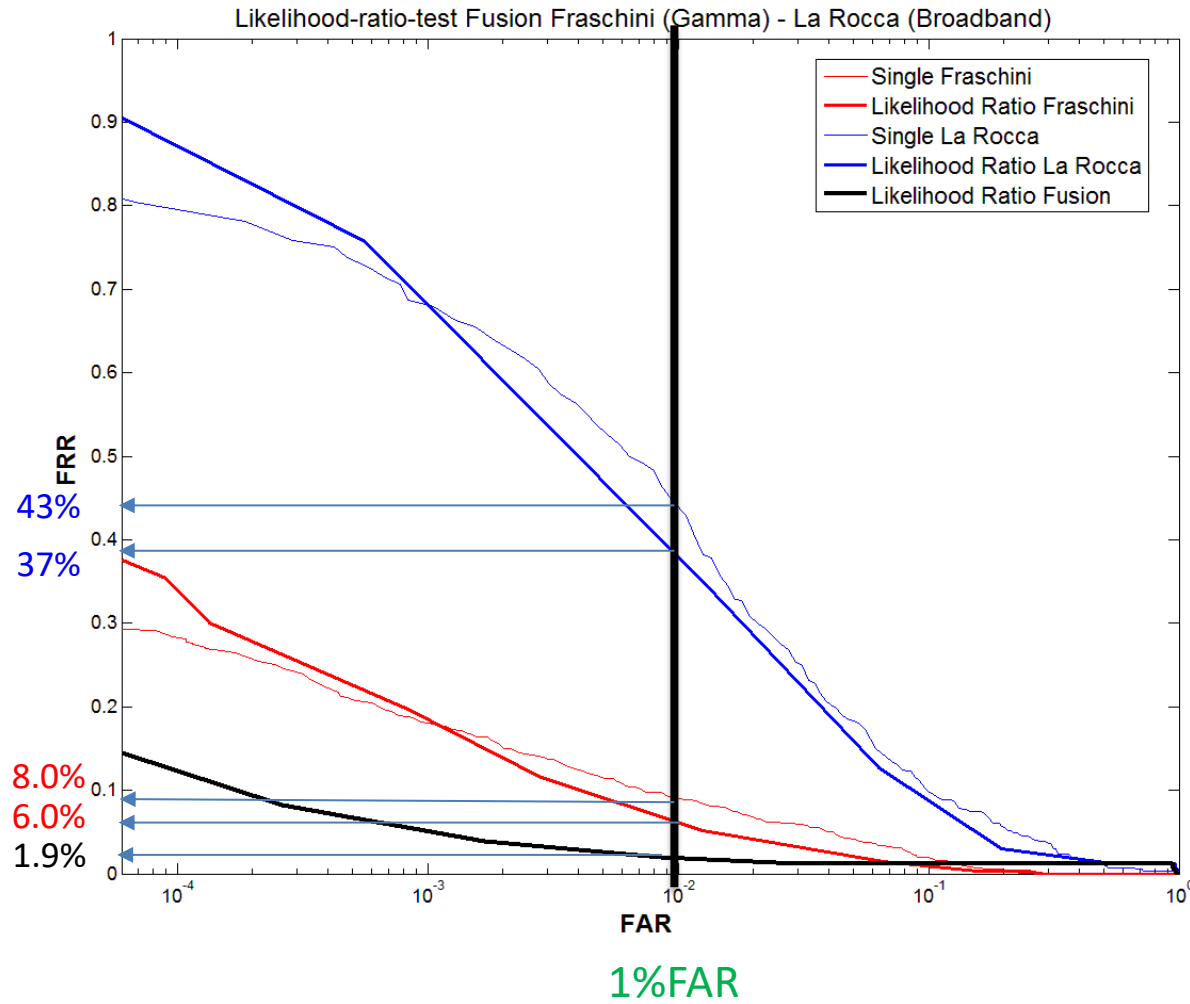
$$conc = (template_{Frasch}, template_{LaRoc})$$

- **Score Level by geometric average (SLF):**

$$s^* = \sqrt{s_{Frasch} * s_{LaRoc}}$$

EER	FLF	SLF
EO	2.98%	1.63%
EC	4.07%	2.35%

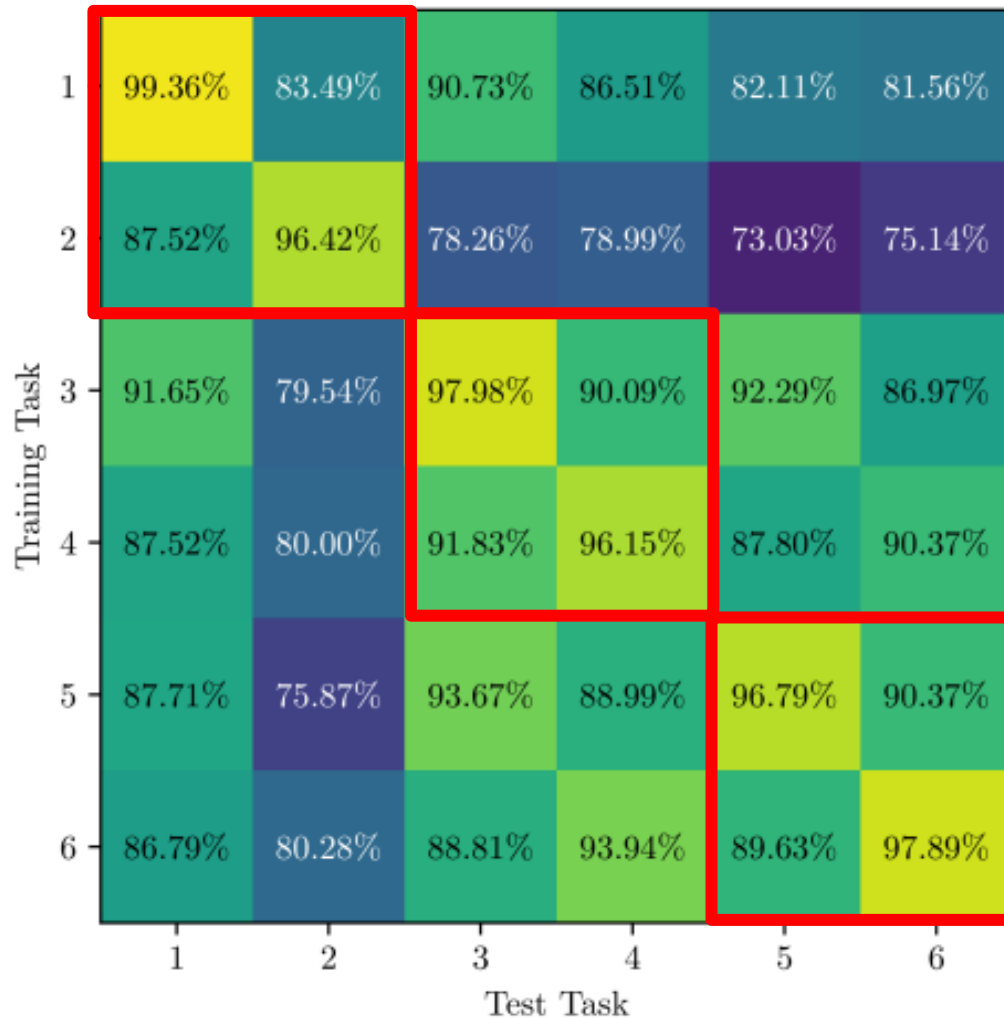
Likelihood ratio test based Fusion



$$LR(x) = \frac{p(s_{Frasc}|gen) * p(s_{LaRoc}|gen)}{p(s_{Frasc}|imp) * p(s_{LaRoc}|imp)}$$

EER	EO
LR Fusion	1.40%
Fraschini-Marcialis	4.46%
La Rocca-Campisi	9.91%

EEG recognition by multiple patch fusion



1. Resting state with eyes open
2. Resting state with eyes closed
3. Open and close left or right fist
4. Imagine opening left or right fist
5. Open and close both fists or both feet
6. Imagine opening or closing both fists or both feet

EEG recognition over years



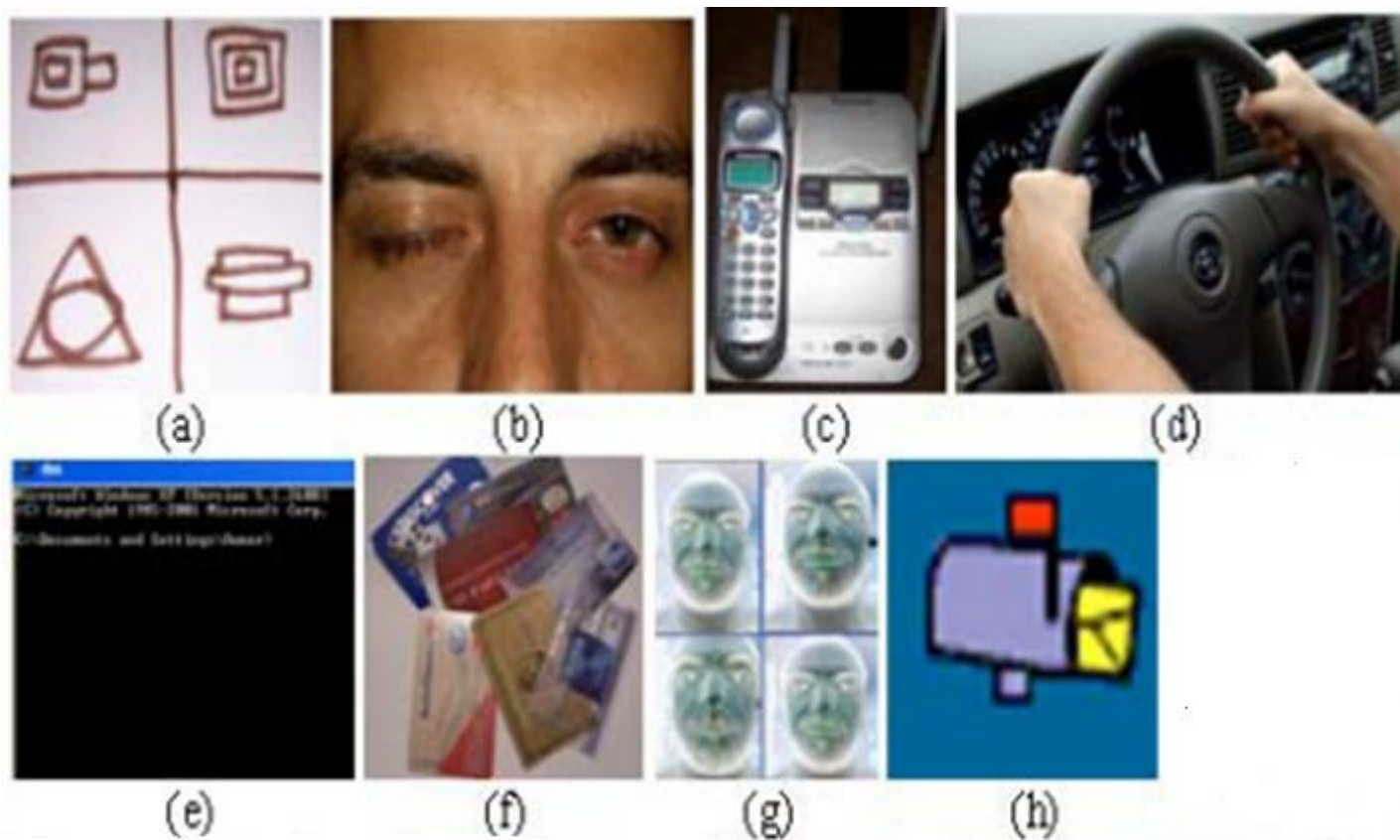
Year	Classification	Dataset	Subjects	Channels	Task type	Accuracy	Epoch
2014	Distance-based	Physionet	108	56	REC	100.00%	10 s
2015	Distance-based	Physionet	109	64	REO	96.90%	12 s
2015	CNN	Physionet	10	64	REO	88.00%	1 s
2016	LDA	Physionet	108	9	REO	96.15%	30 s
2017	CNN	BCIT	100	64	Drive	97.00%	1 s
2017	CNN	-	40	17	VEP	99.80%	0.6 s
2018	CNN	SSVEP	10	-	VEP	95.90%	1 s
2018	CNN/GRU	DEAP	32	32	LL	99.90%	10 s
2019	CNN/LSTM	Physionet	109	16	Multi-tasks	99.58%	1 s
2019	CNN/LSTM	Physionet	109	64	REO	99.95%	12 s
2019	GCNN	Physionet	109	64	REO	99.75%	3 s
2020	Graph-based	Physionet	109	64	REO	98.83%	1 s
2021	CNN	Physionet	109	64	REO	99.91%	1 s
2021	CNN	DEAP	32	32	LL	99.99%	1 s

More behavioural biometrics



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(a) Biometric sketch, (b) blinking, (c) calling, (d) car driving, (e) command line lexicon, (f) credit card use, (g) dynamic facial features, (h) e-mail



More behavioural biometrics



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(j) Game strategy, (k) GUI interaction, (l) handgrip, (m) haptic, (n) keystrokes, (o) lip movement, (p) mouse dynamics, (q) painting style



(j)



(k)



(l)



(m)



(n)



(o)



(p)



(q)

More behavioural biometrics



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(r) Programming style, (s) signature, (t) tapping, (u) text authorship



(r)



(s)



(t)



(u)

That's all for today... what's next?



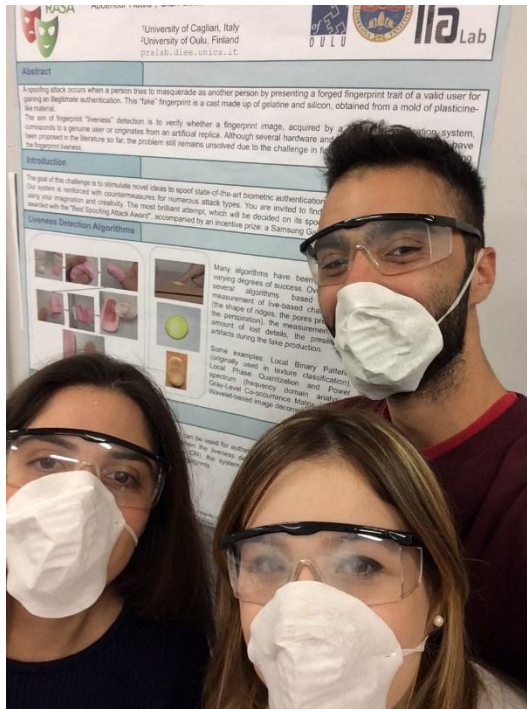
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- From behavioural biometrics to behaviour recognition
- Face expressions
- Gestural analysis
- Anomalous events detection in small and large crowds





Thank you for listening!



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